A Theory of Firm Boundaries and Knowledge Sharing *

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Abstract

We study asset ownership as an instrument for protecting knowledge in collaborations where production requires sharing private knowledge but exposes parties to non-contractible expropriation. Ownership confers control over production and the ability to expropriate. It can protect the owner's knowledge by reorganizing production so certain parties need not be informed, or by ensuring that some parties who must be informed lack the means to expropriate — allowing information to be shared safely. However, because ownership also gives the owner the ability to expropriate, other parties may withhold their knowledge from the owner. We apply the theory to explain the role of third-party intermediaries that facilitate knowledge sharing among competitors, and to determine when a vertically integrated firm must spin off a subsidiary to encourage knowledge sharing from rivals versus when internal "Chinese Walls" suffice.

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1 Introduction

How do firm boundaries shape the transfer and use of knowledge? Recent empirical work in organizational economics documents a regularity: when one firm buys another, acquired firms often come to resemble their acquirers along a range of organizational and operational dimensions (Atalay, Hortaçsu, and Syverson, 2014; Braguinsky et al., 2015; Eliason et al., 2020; Demirer and Karaduman, 2025). These findings are typically interpreted as evidence that integration facilitates the transfer of knowledge or other intangible inputs.

One view, going back to Arrow (1975), is that it is less costly to communicate knowledge within firms than across firm boundaries. As an explanation, however, this view shifts the question rather than answering it, leaving unexplained the mechanism through which integration reduces these communication costs.[1] Instead of taking communication costs as primitive, we focus on incentives to share knowledge. We assume that knowledge can be freely shared regardless of firm boundaries, but that its subsequent use is difficult to control once it is revealed (Arrow, 1962). In particular, sharing knowledge with another party can enable non-contractible follow-on use that benefits the recipient at the expense of the original knowledge holder. We refer to this follow-on use as knowledge expropriation. The core question we study is how firm boundaries shape incentives to share knowledge and, in turn, determine its use in production.

We argue that firm boundaries matter in two related but distinct ways. To see both in a stark setting, consider the following stylized example. There are two master chefs. Each has a secret recipe for a component of a dish. If both recipes are used together, they can produce a breakthrough dish. Recipes can be disclosed freely, but disclosure creates a risk of expropriation: once one chef learns the other's recipe, the recipient can use it outside the collaboration, for example, to launch a competing dish. Following the property-rights tradition (Grossman and Hart, 1986; Hart and Moore, 1990; Hart, 1995), we model a firm boundary choice as allocating ownership — and hence control — over a key productive asset. In this example, suppose the asset is proprietary equipment required to produce the dish.

First, ownership may confer *control over production*. Suppose the dish must be produced by a single chef who owns and operates the equipment. Both chefs can expropriate each other's knowledge, but the owner can deploy their own recipe directly without disclosing it. The dish nevertheless requires both component recipes, so the operating chef must learn the other chef's recipe. Anticipating expropriation risk, the non-owner may be unwilling to fully disclose the

[1] See also Grossman and Hart (1986): "Arrow (1975) has analyzed the benefits of vertical integration based on the assumption that without integration, it is more costly for one firm to communicate information to another than with integration. The incentives of people to lie may change if their incentive structure changes, but Arrow does not explain how integration changes the set of feasible incentive structures." (p. 695)

secret. Swapping ownership changes which chef can act on their knowledge without disclosure and which chef must reveal information and thereby become exposed to expropriation.

Second, ownership may confer *control over expropriation*. In contrast to the previous scenario, suppose production requires mutual disclosure: both chefs must know both recipes to succeed. Crucially, however, assume that expropriation requires the proprietary equipment. In this case, the owner is willing to disclose their recipe, as the non-owner cannot commercialize the knowledge without the asset. The non-owner, by contrast, remains unwilling to fully disclose their secret, as the owner possesses the means to expropriate it. Swapping ownership therefore determines which party can safely disclose their knowledge.

This example illustrates how ownership shapes knowledge use by allocating protection against expropriation. This protection takes two forms. First, ownership can confer *control over production*: the owner's knowledge is protected because it can be applied directly without being disclosed. Second, ownership can confer *control over expropriation*: the owner's knowledge is protected because the non-owner lacks the means to misuse it. These two forms of protection have distinct implications for interpreting empirical patterns. When an acquired firm begins to resemble its acquirer, this pattern is conventionally interpreted as evidence of *knowledge transfer*. However, it may instead reflect *knowledge application*: the acquirer gains control over key decisions and applies its knowledge directly, without necessarily revealing it to the acquired firm.

This example is stylized in two important respects. First, in practice, ownership decisions are shaped not only by bilateral incentives but also by broader industry configurations. For instance, competing firms often share knowledge indirectly through common partners such as manufacturing suppliers (Küpper et al., 2020). Second, firms transact not with monolithic entities but with organizations whose internal structure shapes information flows and the risk of misuse. For example, automakers were reluctant to share sensitive information with the auto supplier Delphi Automotive when it was a subsidiary of their competitor General Motors, prompting GM to spin it off.[2] At the same time, firms sometimes claim that internal informational "Chinese Walls" between a parent and a subsidiary can achieve the same goal by assuring competitors that their knowledge will not reach the parent (Martin and Mickle, 2017; Gawer and Cusumano, 2002; Steinhilber, 2008).

We now turn to a model that lets us study these issues in a unified way. In our model, there are multiple parties who collaborate on projects that require actions to be tailored to underlying information. Actions are complementary in the O-ring sense (Kremer, 1993): a project succeeds

[2]See CNN Money (1999): "Delphi... said the spinoff will allow the company to pursue supply contracts with other automakers, which previously were reluctant to use Delphi because it was a unit of GM". Bradsher (1998) notes that rivals avoided Delphi "for fear that it would share parts designs and other competitive information with G.M."

only if all required actions are chosen correctly. Sharing information improves cooperation, but it creates an incentive problem because disclosed information can be misused through non-contractible expropriation actions. Parties can contract over transfers, asset ownership, and an information-sharing protocol that governs what each party learns prior to choosing actions. We focus on the case in which expropriation is sufficiently costly that optimal contracts deter it.

For a fixed ownership structure, the problem reduces to designing the information structure. Following Bergemann and Morris (2019), we represent this choice as a distribution over private action recommendations. These recommendations must satisfy an obedience constraint: upon receiving a recommendation, each party must find it optimal to follow it given the information the recommendation conveys. In our setting, obedience requires both choosing the recommended productive action and refraining from expropriation.

A convenient way to summarize the no-expropriation constraints is through safety thresholds. Each project requires a collection of tasks: for a given task, one party must act on another party's private information. When the acting party is capable of expropriation, the no-expropriation constraint places an upper bound on the precision of the signal it can receive. We call this bound the task's safety threshold: the greatest precision that can be disclosed to that party about that target's information while ensuring the party refrains from expropriation. The threshold is higher when expropriation is privately less attractive and lower when it is more attractive.

Our first main result is that, for any fixed ownership structure, the optimal information structure has a nested form. Rank the required tasks by their safety thresholds. The optimal contract coordinates recommendations so that whenever a low-threshold ("hard") task is implemented accurately, all higher-threshold ("easier") tasks are implemented accurately as well. Nesting is efficient because tasks are complementary: accuracy on a high-threshold task is wasted in states where a low-threshold task fails. The optimal contract therefore concentrates accuracy in the same states. As a result, project success is constrained by its information bottleneck: the required task with the lowest safety threshold.

We use this characterization to clarify the role of ownership. Ownership ties production control to expropriation capability: the owner must be informed to make productive decisions and can misuse disclosed knowledge. Shifting ownership changes which parties are both required for success and capable of expropriation, and thus changes the information bottlenecks. Ownership can relax these bottlenecks either by disarming the bottleneck party (i.e., the party responsible for the bottleneck task) or by reorganizing production so that this party need not be informed. Our second main result characterizes optimal ownership as the structure that maximizes total surplus, where each project is constrained by its bottleneck and weighted by its value.

We apply these results to several prominent industry configurations. First, we show how

common suppliers and other intermediaries can act as knowledge hubs: when direct sharing among competitors would make a low-threshold rival the bottleneck, routing information through a safer supplier improves cooperation. Second, we rationalize vertical spin-offs as a way to encourage rivals to share knowledge. When an upstream unit is owned by a downstream competitor, outside customers may withhold information for fear it will reach the parent (as in GM's spin-off of Delphi). Separation shifts the bottleneck to a party with weaker incentives to expropriate. Third, we characterize when internal Chinese Walls can substitute for separation. A Chinese Wall is credible only when the rival-facing project can be implemented without operational involvement by the parent — and thus without exposing the parent to rivals' information — in a way that would re-create the bottleneck.

We conclude by considering three variations of the model that clarify our main results. First, we show that ownership matters for knowledge sharing due to the *non-contractibility of expropriation*. If parties could sign a binding no-expropriation agreement before disclosing knowledge, they could achieve the first-best regardless of ownership. By contrast, contracting over productive actions or the division of collaborative profits does not resolve the underlying incentive problem. We view the non-contractibility of expropriation as a fundamental property of knowledge: once disclosed, it cannot be unlearned, and it is generally difficult to verify how it is used outside the collaboration.

Second, we show that asset ownership cannot substitute for contracts because it confers "too many" *inextricable control rights*. An owner always has the means for expropriation and is typically required to make productive decisions using others' information. Reallocating ownership can therefore shift which party becomes the bottleneck, but rarely eliminates it altogether. Separating various control rights across multiple assets would weakly increase welfare and could achieve the first-best. We view inseparability of various control rights as a fundamental limitation of integration: ownership reallocations typically transfer a coarse bundle of *residual* rights (Grossman and Hart, 1986), not the targeted control needed for efficient use of knowledge.

Finally, we discuss the role of expropriation costs. In the baseline model, expropriation is sufficiently costly that optimal contracts deter it: temptation to expropriate shapes information disclosure, but expropriation costs are never incurred. When expropriation is costly but not prohibitive, it can be optimal to tolerate some expropriation to enable greater disclosure and raise the probability of successful cooperation. As a result, optimal ownership depends not only on temptation to expropriate but also on the level of expropriation costs. We characterize the resulting information-design problem via a connection to optimal transport theory.

Literature. We build on the incomplete-contracts approach to firm boundaries in which ownership is modeled as residual control rights (Grossman and Hart, 1986; Hart and Moore,

1990; Hart, 1995). In contrast to the classic property-rights models, inefficiencies are ex post and arise from ex post non-contractible actions (Hart and Holmström, 2010; Legros and Newman, 2013). Control over these actions matters in our model because it constrains who needs to know what and who is willing to share their knowledge with whom.

Within the firm-boundaries literature, our paper is closest methodologically to work that studies boundary choices through a mechanism-design approach (Matouschek, 2004; Baliga and Sjöström, 2018). Whereas these papers emphasize how ownership affects information rents when ex post actions are contractible, we study optimal ownership when ex post actions are non-contractible. The central issue is therefore about obedience rather than truth-telling. To characterize the optimal information-sharing and ownership arrangement in this environment, we use modern information-design tools (Rayo and Segal, 2010; Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019; Mathevet, Perego, and Taneva, 2020).

Finally, we also connect to the literature on communication and delegation in organizations (Dessein, 2002; Alonso, Dessein, and Matouschek, 2008; Rantakari, 2008), which studies how the allocation of control shapes communication. Because ours is a paper on firm boundaries rather than internal organization, we emphasize two features of the environment that delegation models typically abstract from: parties have access to rich contracting instruments (i.e., they can contract jointly on decision rights, information-sharing protocols, and transfers), and ownership allocates a bundle of inextricable control rights.

2 Model

There are N parties, $i \in \mathcal{N} = \{1, \ldots, N\}$, who collaborate on a collection of projects. Each project succeeds only if a set of required actions are correctly tailored to private information held by the parties. Sharing information improves cooperation but creates a risk: parties may use what they learn to expropriate others' knowledge for their private benefit.

Information. The state is $\theta = (\theta_1, \ldots, \theta_N)$, where $\theta_i \in \{-1, 1\}$ is privately observed by party i. The components of θ are independent, and each realization is equally likely, $p(\theta) = \left(\frac{1}{2}\right)^N$.

Actions and Asset Ownership. Each party $i \in \mathcal{N}$ chooses a vector of non-contractible actions. We distinguish actions along two dimensions: (i) whether they are *cooperative* or *expropriation* actions and (ii) whether they are *alienable* or *inalienable*. Cooperative actions are potentially productive. When correctly matched to the relevant component of the state, they contribute to project success. Expropriation actions are privately beneficial but socially destructive. If correctly matched to others' information, they generate a private gain for the acting party and impose a cost on others. Alienable actions require access to a productive asset. Inalienable actions reflect parties' specific human capital and thus do not require access to a

productive asset.

There is a single productive asset owned by party $g \in \mathcal{N}$. We model control over alienable actions in the following way. Each party $i \in \mathcal{N}$ selects a plan for *alienable cooperative actions*, denoted $x_{ij} \in \{-1, 1\}$ for all $j \in \mathcal{N}$, and for *alienable expropriation actions*, denoted $z_{ij} \in \{-1, 0, 1\}$ for all $j \neq i$. While every party specifies a plan, only the choices of the asset owner g can be payoff-relevant.

Each party $i \in \mathcal{N}$ also chooses *inalienable cooperation actions*, denoted by $\bar{x}_{ij} \in \{-1, 1\}$ for all $j \in \mathcal{N}$, and *inalienable expropriation actions*, denoted by $\bar{z}_{ij} \in \{-1, 0, 1\}$ for $j \neq i$.

For each party $i \in \mathcal{N}$, let $a_i = \left((x_{ij})_{j\in\mathcal{N}}, (\bar{x}_{ij})_{j\in\mathcal{N}}, (z_{ij})_{j\neq i}, (\bar{z}_{ij})_{j\neq i}\right)$ denote party i's *action plan*. We denote by $\mathcal{A}_i$ the set of possible action plans by party i and write $\mathcal{A} = \prod_{i\in\mathcal{N}} \mathcal{A}_i$ and $a = (a_i)_{i\in\mathcal{N}} \in \mathcal{A}$. Equivalently, we may write $a = (x, \bar{x}, z, \bar{z})$.

Some parties can expropriate only if they own the asset, while others can expropriate even if they do not. Let $\mathcal{E} \subseteq \mathcal{N}$ denote the set of parties whose expropriation opportunities are tied to the asset. We define party i's effective expropriation action against party j as ζ_{ij}. For those parties who can intrinsically expropriate, $i \notin \mathcal{E}$, $\zeta_{ij} = \bar{z}_{ij}$. For those parties in $i \in \mathcal{E}$, $\zeta_{ij} = z_{ij}$ for the asset owner ($g = i$) and $\zeta_{ij} = 0$ for all others ($g \neq i$).

Projects. Production is organized into a set of projects $\mathcal{K}$. Each project $k \in \mathcal{K}$ has value $V_k > 0$ if successfully implemented and value 0 otherwise. Project success requires that a set of cooperative actions be correctly matched to the relevant components of the state.

Formally, each project k has two sets of requirements. First, $\mathcal{R}_k \subseteq \mathcal{N}$ is the set of alienable requirements: for each $j \in \mathcal{R}_k$, the asset owner g must set $x_{gj} = \theta_j$. Second, $\bar{\mathcal{R}}_k \subseteq \mathcal{N} \times \mathcal{N}$ is the set of inalienable requirements: for each $(i, j) \in \bar{\mathcal{R}}_k$, party i must choose $\bar{x}_{ij} = \theta_j$. Given asset owner g, define the project-success indicator $w_k \in \{0, 1\}$ by

$$w_k(x, \bar{x}; g, \theta) = \left(\prod_{j\in\mathcal{R}_k} \mathbf{1}\{x_{gj} = \theta_j\}\right)\left(\prod_{(i,j)\in\bar{\mathcal{R}}_k} \mathbf{1}\{\bar{x}_{ij} = \theta_j\}\right).$$

Project k succeeds if and only if all of its alienable and inalienable requirements are met.

Payoffs. Party i's payoffs are quasilinear and consist of three components:

$$u_i(a, t; g, \theta) = u_i^c(x, \bar{x}; g, \theta) + u_i^e(z, \bar{z}; g, \theta) + t_i,$$

where u_i^c is the *cooperative payoff* from project success, u_i^e is the *expropriation payoff*, and t_i is a monetary transfer. Transfers are budget balanced, so $\sum_{i\in\mathcal{N}} t_i = 0$. We next describe the cooperative and expropriation payoff terms.

The cooperative payoff depends only on which projects succeed. We assume party i receives

an exogenous share $\alpha_{ik} \geqslant 0$ of project k's value, with $\sum_{i \in \mathcal{N}} \alpha_{ik} = 1$. Its cooperative payoff is

$$u_i^c(x, \bar{x}; g, \theta) = \sum_{k \in \mathcal{K}} \alpha_{ik} V_k w_k(x, \bar{x}; g, \theta).$$

Thus, cooperative payoffs depend on actions only through project success and do not depend directly on asset ownership beyond its effect on which actions are payoff-relevant.

Expropriation payoffs depend on the effective expropriation actions ζ_{ij}. If party i attempts to expropriate party j's knowledge, and the attempt is successful (i.e., $\zeta_{ij} = \theta_j$), then party i obtains a private benefit $b_{ij} > 0$, while party j incurs a loss $\lambda_{ij} > 0$. If the attempt is unsuccessful (i.e., $\zeta_{ij} = -\theta_j$), party i incurs a cost $c_{ij} > 0$. If $\zeta_{ij} = 0$, both parties receive 0. Party i's expropriation payoff is therefore

$$u_i^e(z, \bar{z}; g, \theta) = \sum_{j \neq i} \left[b_{ij} 1\{\zeta_{ij} = \theta_j\} - c_{ij} 1\{\zeta_{ij} = -\theta_j\} \right] - \sum_{j \neq i} \lambda_{ji} 1\{\zeta_{ji} = \theta_i\}.$$

We define total surplus as $TS(a; g, \theta) = \sum_{i \in \mathcal{N}} u_i(a, t; g, \theta)$. Because transfers are budget-balanced, they do not directly enter this expression.

We impose the following two assumptions on expropriation payoff:

Assumption 1. *For each party $i \in \mathcal{N}$ and target $j \neq i$, $\lambda_{ij} - b_{ij} > V_{\text{total}} \equiv \sum_{k \in \mathcal{K}} V_k$.*

Assumption 2. *For each party $i \in \mathcal{N}$ and target $j \neq i$, $b_{ij} < c_{ij}$.*

The first assumption ensures any successful expropriation destroys more total surplus than can be generated by all projects. The second assumption ensures expropriation is risky and thus attractive only when a party has sufficiently precise information about the target's state.

Solution Concept. Consider an *omniscient* mediator who decides how information is disclosed to the parties. A *decision rule* is a mapping

$$\sigma(\cdot \mid \theta) \in \Delta(\mathcal{A}) \qquad \text{for each } \theta \in \Theta,$$

which assigns to each realized state θ a distribution over action profiles $a = (a_i)_{i \in \mathcal{N}}$.

The timing is as follows: (i) the mediator commits to a decision rule σ; (ii) the state θ is realized; (iii) each party i observes θ_i; (iv) the mediator draws an action profile a according to $\sigma(\cdot \mid \theta)$ and privately recommends a_i to each party i; (v) the parties choose their actions; and (vi) payoffs are realized.

Recommendations are not enforceable. Instead, a decision rule must be designed so that parties optimally choose to follow their recommendations given their information. Accordingly, we restrict attention to *obedient decision rules*, under which no party wishes to deviate from the

recommended action after observing its private recommendation and private state. We denote the set of obedient recommendation rules by Σ. Formally, $\sigma \in \Sigma$ if

$$\sum_{\theta_{-i}\in\Theta_{-i}} \sum_{a_{-i}\in\mathcal{A}_{-i}} p(\theta_{-i}|\theta_i)\sigma(a_i, a_{-i} \mid \theta_i, \theta_{-i})\Big[u_i\big((a_i, a_{-i}), g; \theta_i, \theta_{-i}\big) - u_i\big((a_i', a_{-i}), g; \theta_i, \theta_{-i}\big)\Big] \geqslant 0$$
$$\text{for all } i \in \mathcal{N},\ \theta_i \in \Theta_i,\ a_i \in \mathcal{A}_i,\ a_i' \in \mathcal{A}_i. \tag{2.1}$$

This formulation follows the information-design-in-games approach of Bergemann and Morris (2019). An obedient decision rule σ induces a joint distribution over states and actions, $\psi(a, \theta) = p(\theta)\sigma(a \mid \theta)$. The decision rule is Bayes-plausible, $\sum_a \psi(\theta, a) = p(\theta)$, and obedience corresponds to the requirement that the induced distribution constitutes a Bayes correlated equilibrium, as captured by (2.1).

We study *optimal decision rules* that maximize total surplus subject to obedience. Let

$$TS(g) \equiv \max_{\sigma\in\Sigma} \mathbb{E}\left[TS(a, g; \theta)\right], \tag{2.2}$$

where the expectation is taken with respect to the distribution $\psi(a, \theta) = p(\theta)\sigma(a \mid \theta)$. *Optimal asset ownership* is g^* that solves $\max_{g\in\mathcal{N}} TS(g)$.

Discussion of Modeling Assumptions. The model relies on several assumptions that we relax and discuss in the extensions and appendix.

First, our solution concept assumes an omniscient mediator who can condition recommendations on the full state θ. We justify this formulation as a reduced form of a general contracting game. Suppose parties can contract over (i) (possibly random) asset ownership, (ii) monetary transfers, and (iii) the behavior of a neutral mediator. A contract is a *coordination mechanism* in the sense of Myerson (1982, 1983): parties send messages to the mediator, the mediator enforces the contractible components (ownership and transfers), sends recommendations, and parties then choose non-contractible actions. A feasible mechanism must satisfy several incentive constraints. In addition to obedience described above, it must satisfy truth-telling (parties report their states truthfully), rule out profitable double deviations that combine misreporting with disobedience, and satisfy participation constraints ensuring that each party prefers acceptance to the disagreement payoff. In Appendix A.2, we show that the solution to the full mechanism-design problem coincides with the solution concept in (2.2). Viewed this way, the omniscient mediator formulation provides a convenient characterization of the outcomes induced under optimal mechanisms.

Second, we assume that parties receive exogenous shares of project values, $\{\alpha_{ik}\}_{i,k}$. Section 5.1 allows these shares and cooperative actions to be contractible and shows that doing so does not increase the maximum attainable surplus.

Third, the baseline model assumes a single productive asset that bundles control over all alienable actions. Section 5.2 allows for multiple assets and for cooperative and expropriation control rights to be allocated separately. This extension clarifies how bundling control in a single asset restricts the maximum attainable surplus.

Finally, we assume that successful expropriation destroys total surplus. Assumption 1 implies that expropriation is never socially optimal and therefore does not occur on the equilibrium path, even though the threat of expropriation shapes information sharing and ownership. This assumption is stronger than necessary for our main results, and the proof of Proposition 1 derives a weaker condition. In addition, Section 5.3 discusses the model under arbitrary expropriation costs and connects the mediator's problem to optimal transport theory.

3 Analysis

The fundamental constraint on total surplus is the threat of expropriation. Party i can profitably attempt to expropriate party j's knowledge only if it simultaneously has (i) the *means* to expropriate (access to an effective expropriation action under the current ownership structure) and (ii) sufficiently precise *information* about θ_j. As we show below, it is optimal to ensure that no party is both *armed* (has expropriation means) and sufficiently *well-informed* about a target to engage in expropriation.

This goal can be achieved through two channels. First, *information design*: holding ownership fixed, any party that is both armed and required for a project's success must receive capped information (up to its safety threshold), while parties without expropriation means can be fully informed safely. Second, *ownership and production organization*: changing ownership affects who has asset-dependent expropriation means and who must be informed for production. Thus, expropriation can be deterred either by removing a party's means of expropriation or by structuring production so it need not learn others' information.

3.1 Optimal Information Disclosure

Fix an owner $g \in \mathcal{N}$ and an obedient decision rule $\sigma(\cdot \mid \theta) \in \Delta(\mathcal{A})$. After observing private state θ_i and recommendation a_i, party i forms posterior beliefs about others' states. For $j \neq i$, let

$$\tau_{ij}(\theta_i, a_i) \equiv \mathbb{P}_\psi\big(\theta_j = 1 \,|\, \theta_i, a_i\big),$$

where the posterior is defined under the joint distribution $\psi(\theta, a) = p(\theta)\sigma(a|\theta)$. When the dependence on (θ_i, a_i) is clear from context, we write τ_{ij} for $\tau_{ij}(\theta_i, a_i)$.

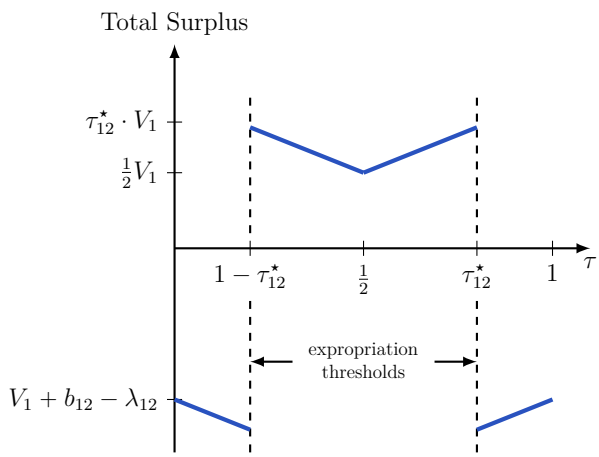

Figure 1: Total Surplus As a Function of Beliefs (Example 1)

This figure illustrates the relationship between total expected surplus and party 1's belief that $\theta_2 = 1$ for Example 1.

If party i can engage in expropriation, it will choose $\zeta_{ij} = 1$ if $\tau_{ij} b_{ij} - (1 - \tau_{ij}) c_{ij} > 0$ and $\zeta_{ij} = -1$ if $(1 - \tau_{ij}) b_{ij} - \tau_{ij} c_{ij} > 0$. Combining these inequalities, expropriation occurs if and only if the belief is sufficiently precise:

$$\max\{\tau_{ij}, 1 - \tau_{ij}\} > \tau^\star_{ij} \equiv \frac{c_{ij}}{b_{ij} + c_{ij}} > \frac{1}{2},$$

where we assume that in the case of indifference, parties do not expropriate.

Before stating the general results on the structure of the optimal decision rule, it is useful to build intuition with two examples.

Example 1: Optimal Decision Rule For One Party. Suppose there are two parties, $\mathcal{N} = \{1, 2\}$, and one project, $\mathcal{K} = \{1\}$. The project does not require an asset, $\mathcal{R}_1 = \varnothing$, but requires party 1 to match the state of party 2 via its inalienable action, $\bar{\mathcal{R}}_1 = \{(1, 2)\}$. That is, the project succeeds if and only if $\bar{x}_{12} = \theta_2$. Parties do not require the asset to engage in expropriation, $\mathcal{E} = \varnothing$. In this example, the asset ownership is irrelevant. Suppose both parties receive half of the project's value, $\alpha_{11} = \alpha_{21} = 1/2$.

The optimal decision rule can be found using the techniques from Kamenica and Gentzkow (2011). Specifically, denote by τ the posterior belief of party 1 that $\theta_2 = 1$ after receiving the recommendation from the mediator. Figure 1 illustrates the expected total surplus as a function τ. It is optimal for party 1 to select $\bar{x}_{12} = 1$ if $\tau > 1/2$ and $\bar{x}_{12} = -1$ if $\tau < 1/2$. If party 1's information about θ_2 is sufficiently precise, it also optimally engages in expropriation. Specifically, If $\tau > \tau^\star_{12}$, party 1 selects $\zeta_{12} = 1$, and $\tau < 1 - \tau^\star_{12}$, party 1 selects $\zeta_{12} = -1$.

Under expropriation, the total expected surplus is negative. Thus, the optimal decision rule

is to randomize between posteriors $\tau^\star_{12}$ and $1-\tau^\star_{12}$. It can be constructed as follows. Suppose there is a random variable $\xi \sim \text{Unif}[0,1]$ independent of θ. The recommendations to party 1 are

$$x_{12} = \bar{x}_{12} = \begin{cases} \theta_2 & \text{if } \xi \leqslant \tau^\star_{12} \\ -\theta_2 & \text{if } \xi > \tau^\star_{12} \end{cases} \quad \text{and} \quad z_{12} = \bar{z}_{12} = 0$$

Example 2: Nested Decision Rule For Multiple Parties. Suppose there are three parties, $\mathcal{N} = \{1,2,3\}$, and two projects, $\mathcal{K} = \{1,2\}$. Neither project has alienable requirements, so $\mathcal{R}_1 = \mathcal{R}_2 = \varnothing$. Project 1 requires two inalienable actions from parties 1 and 2 to match the state of party 3, so $\bar{\mathcal{R}}_1 = \{(1,3),(2,3)\}$. Project 2 requires only an inalienable action from party 2 to match the state of party 3, so $\bar{\mathcal{R}}_2 = \{(2,3)\}$. All parties possess inalienable means of expropriation ($\mathcal{E} = \varnothing$), so asset ownership plays no role in this example.

Suppose expropriation parameters are $b_{13} = 2$, $c_{13} = 3$, $b_{23} = 1$, and $c_{23} = 9$. Then

$$\tau^\star_{13} = \frac{3}{2+3} = 0.6 \qquad \text{and} \qquad \tau^\star_{23} = \frac{9}{1+9} = 0.9.$$

To deter expropriation, party i's recommendation for the action $\bar{x}_{i3}$ must be such that its posterior belief τ_{i3} about θ_3 lies in the safety region $[1-\tau^\star_{i3}, \tau^\star_{i3}]$. Following Example 1, a natural obedient rule is to recommend the correct match to party i with probability $\tau^\star_{i3}$ and the incorrect match with probability $1-\tau^\star_{i3}$. However, with two parties required for Project 1, the key question is whether these recommendations should be generated independently.

If parties 1 and 2 receive independent recommendations satisfying their respective safety thresholds, then Project 1 succeeds with probability $\tau^\star_{13} \cdot \tau^\star_{23}$, while Project 2 succeeds with probability $\tau^\star_{23}$. Hence the expected total surplus is

$$V_1 \cdot \tau^\star_{13} \cdot \tau^\star_{23} + V_2 \cdot \tau^\star_{23} = V_1 \cdot 0.54 + V_2 \cdot 0.9$$

The mediator can do strictly better by *positively correlating* recommendations. Intuitively, party 1 is the bottleneck in Project 1, so whenever party 1 is recommended the correct action, party 2 should also be recommended the correct action.

Let $\xi \sim \text{Unif}[0,1]$ be a *common* randomization device. Consider the nested rule in which, for each $i \in \{1,2\}$, the mediator recommends $\bar{x}_{i3} = \theta_3$ if $\xi \leqslant \tau^\star_{i3}$ and $\bar{x}_{i3} = -\theta_3$ otherwise. This rule is illustrated in Figure 2. Under it, Project 1 succeeds with probability $\min\{\tau^\star_{13}, \tau^\star_{23}\} = \tau^\star_{13}$, while Project 2 succeeds with probability $\tau^\star_{23}$. Thus, expected total surplus is

$$V_1 \cdot \min\{\tau^\star_{13}, \tau^\star_{23}\} + V_2 \cdot \tau^\star_{23} = V_1 \cdot 0.6 + V_2 \cdot 0.9,$$

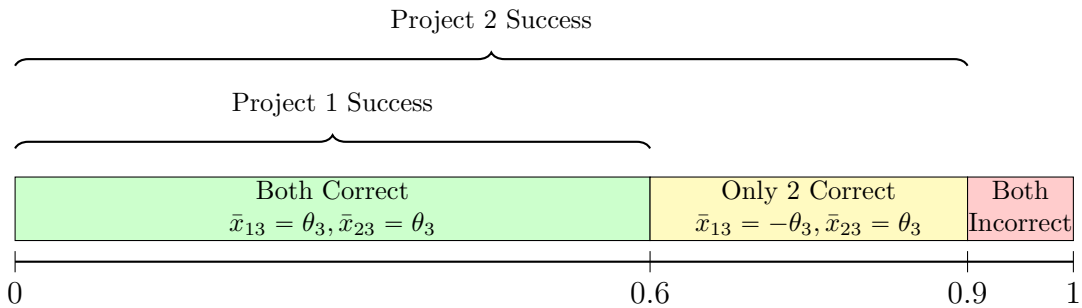

Figure 2: Nested Decision Rule For Example 3

The figure illustrates the realization of recommendations based on the random variable $\xi \sim \text{Unif}[0,1]$. Party 1 receives the correct recommendation only when $\xi \leqslant 0.6$. Party 2 receives the correct recommendation when $\xi \leqslant 0.9$. This decision rule ensures that Party 2 is always informed whenever Party 1 is informed.

which strictly exceeds the independent benchmark. Proposition 1 below shows that such nested decision rules — implemented via a common randomization device — are optimal more generally.

General Results. Fix an owner $g \in \mathcal{N}$. Let $\text{Arm}(g)$ denote the set of "armed" parties, that is parties, who can engage in expropriation under owner $g \in \mathcal{N}$:

$$\text{Arm}(g) = \{i \in \mathcal{N} : i = g \text{ or } i \notin \mathcal{E}\}.$$

This set includes the owner and all parties with inalienable means for expropriation ($i \notin \mathcal{E}$).

Disarmed parties can be fully informed about others because they cannot expropriate. For armed parties, by contrast, information must be capped at the expropriation threshold. Let $\tau_{ij}^{\text{safe}}(g)$ denote the maximum precision of party i's belief about θ_j that is consistent with no expropriation:

$$\tau_{ij}^{\text{safe}}(g) = \begin{cases} \tau_{ij}^{\star} = \frac{c_{ij}}{b_{ij}+c_{ij}} & \text{if } i \in \text{Arm}(g) \\ 1 & \text{if } i \notin \text{Arm}(g) \end{cases}.$$

For convenience, we set $\tau_{ii}^{\text{safe}}(g) \equiv 1$ for all $i \in \mathcal{N}$ and $g \in \mathcal{N}$.

For project k, it is useful to represent the production requirements as *matching tasks* of the form "party i must match party j's state." Let

$$\mathcal{Q}_k(g) \equiv \{(g,j) : j \in \mathcal{R}_k\} \ \cup \ \bar{\mathcal{R}}_k \ \subseteq \ \mathcal{N} \times \mathcal{N}$$

denote the set of such tasks that must be performed for project k to succeed under ownership g.

The next proposition establishes that for each project $k \in \mathcal{K}$ the maximum attainable expected surplus is determined by the *bottleneck task*: the lowest safety threshold $\tau_{ij}^{\text{safe}}(g)$ among tasks in $(i,j) \in \mathcal{Q}_k(g)$.

Proposition 1 (Optimal Decision Rule)**.** *Fix an asset owner g. Under Assumptions 1–2, the maximum attainable expected total surplus is*

$$TS(g) = \sum_{k \in \mathcal{K}} V_k \cdot \left(\min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) \right). \tag{3.1}$$

Moreover, there exists an optimal obedient decision rule with a nested structure: whenever a task (i, j) with lower safety threshold succeeds, all tasks with higher safety thresholds succeed as well. The mediator draws a single common random variable $\xi \sim \text{Unif}[0, 1]$ and, conditional on (θ, ξ), privately recommends:

(i) No expropriation: each party i is recommended to choose $z_{ij} = \bar{z}_{ij} = 0$ for all $j \neq i$.

(ii) Cooperation: party i is recommended the correct match ($x_{ij} = \bar{x}_{ij} = \theta_j$) if and only if $\xi \leqslant \tau_{ij}^{\text{safe}}(g)$, and is recommended the opposite match otherwise ($x_{ij} = \bar{x}_{ij} = -\theta_j$).

Proof. See Section A.1 in the appendix. □

The intuition behind Proposition 1 is as follows. The mediator prefers to avoid expropriation because it destroys total surplus. To do so, information must be partially withheld from *armed* parties — those who have the means to expropriate. Therefore, the planner discloses information to each party i about state θ_j only up to its safety threshold $\tau_{ij}^{\text{safe}}(g)$.

Since cooperative actions are complementary, the probability of successful cooperation in a project is bounded the bottleneck task: the party-target pair with the lowest safety threshold, captured by the term $\min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g)$. The nested decision rule described in the proposition achieves this bound by correlating agents' information through a common randomization device. This decision rule creates a strict hierarchy of information: if a less trustworthy agent is correctly informed about a particular state, then all more trustworthy agents are necessarily informed about this state as well.

3.2 Optimal Asset Ownership

Proposition 1 characterizes the maximum total surplus as a function of ownership structure $g \in \mathcal{N}$. Optimal asset ownership solves the following problem

$$g^* \in \arg\max_{g \in \mathcal{N}} TS(g) = \arg\max_{g \in \mathcal{N}} \left\{ \sum_{k \in \mathcal{K}} V_k \cdot \left(\min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) \right) \right\}.$$

In this subsection, we derive several results on how the optimal asset ownership depends on the model's primitives. To facilitate the discussion, let $\mathcal{P}_k(g)$ denote the set of *active* participants

in project k under ownership $g \in \mathcal{N}$:

$$\mathcal{P}_k(g) \equiv \left\{ i \in \mathcal{N} : \text{there exists some } j \neq i \text{ such that } (i,j) \in \mathcal{Q}_k(g) \right\}. \tag{3.2}$$

That is, $\mathcal{P}_k(g)$ consists of parties who must match the state of some *other* party for project k to succeed.

We start with two examples that clarify the two roles of ownership. First, control over production operates through the task set $\mathcal{Q}_k(g)$ by determining the set of active participants $\mathcal{P}_k(g)$. Second, control over expropriation operates through the safety thresholds $\tau_{ij}^{\text{safe}}(g)$ by determining who has the means to expropriate.

Example 3: Ownership as Control Over Production. Suppose there are two parties, $\mathcal{N} = \{1,2\}$, and a single project, $\mathcal{K} = \{1\}$. The project requires an asset owner to match the states of both parties to succeed, so $\mathcal{R}_1 = \{1,2\}$. No inalienable actions are required ($\bar{\mathcal{R}}_1 = \varnothing$), and all parties possess the inalienable means to expropriate ($\mathcal{E} = \varnothing$).

Given an owner $g \in \{1,2\}$, the project succeeds if and only if $x_{g1} = \theta_1$ and $x_{g2} = \theta_2$. The owner can always match its own state, $x_{gg} = \theta_g$, without disclosing it. By contrast, the non-owner's state must be disclosed to the owner for the project to succeed. Since the owner is armed, the information disclosed about the non-owner's state must be capped to deter expropriation.

Formally, ownership affects the set of active participants, $\mathcal{P}_1(g) = \{g\}$. However, it does not affect safety thresholds ($\tau_{ij}^{\text{safe}}(g) = \tau_{ij}^{\star} = \frac{c_{ij}}{b_{ij}+c_{ij}}$) because expropriation is inalienable ($\mathcal{E} = \varnothing$). The asset should be owned by the party with the weaker temptation to expropriate (equivalently, the higher safety threshold):

$$g^* \in \arg\max_{g \in \mathcal{N}} \left\{ \tau_g^{\star} \cdot V_1 \right\} \quad \Leftrightarrow \quad g^* \in \arg\min_{g \in \mathcal{N}} \left\{ \frac{b_g}{c_g} \right\}, \tag{3.3}$$

where $\tau_g^{\star} \equiv \tau_{gj}^{\star}$, $b_g \equiv b_{gj}$, $c_g \equiv c_{gj}$ for $j \neq g$.

Example 4: Ownership as Control Over Expropriation. Suppose again there are two parties, $\mathcal{N} = \{1,2\}$, and a single project, $\mathcal{K} = \{1\}$. The project requires an asset owner to match the states of both parties ($\mathcal{R}_1 = \{1,2\}$). Unlike the previous case, cooperation now requires complementary inalienable actions: each party must use their human capital to match the other party's state ($\bar{\mathcal{R}}_1 = \{(1,2),(2,1)\}$). Furthermore, we assume that both parties require the asset to engage in expropriation ($\mathcal{E} = \{1,2\}$).

There are two key differences from Example 3. First, cooperation now requires inalienable actions by both parties, and therefore ownership does not affect the set of active participants, $\mathcal{P}_1(g) = \{1,2\}$ for all $g \in \mathcal{N}$. Second, expropriation is alienable and controlled by the owner: the non-owner is disarmed and has no effective expropriation action. Therefore, ownership affects

safety thresholds, $\tau_{ij}^{\text{safe}}(g) = \tau_{ij}^{\star}$ if $i = g$ and $\tau_{ij}^{\text{safe}}(g) = 1$ if $i \neq g$.

As in the previous example, the optimal asset owner is the party with the lowest temptation to expropriate (see (3.3)). The logic, however, differs across the two settings. In Example 3, ownership determines who participates in production. In Example 4, ownership determines who lacks the means to expropriate.

Additional Results. Next, we characterize the optimal ownership structure for several special cases. Let $\mathcal{Q}_{k,i}(g) \equiv \{j \in \mathcal{N} : (i,j) \in \mathcal{Q}_k(g)\}$ denote the set of targets party i must match for project k to succeed. Let $\tau_{k,i}^{\min}(g) \equiv \min_{j \in \mathcal{Q}_{k,i}(g)} \tau_{ij}^{\text{safe}}(g)$ denote party i's bottleneck in project k (the "hardest" task). For notational convenience, we set $\min \varnothing \equiv 1$.

For the next result, we make the following assumption (in addition to Assumptions 1–2):

Assumption 3. *For every ownership structure $g \in \mathcal{N}$ and project $k \in \mathcal{K}$, non-owner active participants are disarmed: $\mathcal{P}_k(g) \cap \text{Arm}(g) \subseteq \{g\}$.*

Under this assumption, any project in which the owner is *not* an active participant can be implemented with certainty: all of its active participants are disarmed and can safely be fully informed. The only projects that are "throttled" by an information cap are those in which the owner must act. Thus, the formula for maximum attainable surplus can be simplified.

Corollary 1. *Under Assumption 3, the maximum attainable surplus under an owner $g \in \mathcal{N}$ is*

$$TS(g) = V_{\text{total}} - \sum_{k \in \mathcal{K}} V_k \cdot \mathbf{1}\{g \in \mathcal{P}_k(g)\} \cdot (1 - \tau_{k,g}^{\min}(g)). \tag{3.4}$$

Proof. Follows directly from the formula in (3.1) under Assumption 3 and the definitions of $\mathcal{P}_k(g)$ and $\tau_{k,g}^{\min}(g)$. □

This corollary generalizes the logic of Examples 3 and 4. Specifically, if there is only one project, and an asset owner is the only armed active participant, then $TS(g) = \tau_{gj}^{\star} \cdot V_1$ for $j \neq g$, and the asset should be assigned to the party with the lowest temptation to expropriate. More generally, suppose there are many projects, and an asset owner is an active participant in all projects ($g \in \mathcal{P}_k(g)$ for all $k \in \mathcal{K}$). Then

$$TS(g) = \sum_{k \in \mathcal{K}} V_k \cdot \tau_{k,g}^{\min}(g),$$

so the optimal ownership trades off project values against the owner's worst bilateral safety threshold toward the targets it must match in each project.

The corollary also shows that the optimal ownership depends on parties' *involvement* in projects. Specifically, suppose that all parties have the same worst bilateral safety thresholds:

$\tau_{k,g}^{\min}(g) \equiv \tau_k^{\min}$ for all $g \in \mathcal{N}$. Then, the optimal owner is

$$g^* \in \arg\min_{g \in \mathcal{N}} \sum_{k \in \mathcal{K}} V_k \cdot \mathbf{1}\{g \in \mathcal{P}_k(g)\} \cdot (1 - \tau_k^{\min}),$$

which depends on g only through $\mathbf{1}\{g \in \mathcal{P}_k(g)\}$. Intuitively, if the owner is the only armed participant, ownership should not be assigned to a party whose human capital (inalienable actions) is required in many high-value projects. Making such a highly involved party the owner would "arm" it, necessitating restricted information flows across all those projects. Assigning ownership to the party who is *least* involved across projects can increase total surplus by allowing the more involved party to be fully informed.

In the next corollary, we focus on ownership as a tool for reallocating the means of expropriation by assuming it does not affect the production requirements.

Corollary 2. *Suppose that for all projects ownership does not affect the production requirements ($\mathcal{Q}_k(g) = \mathcal{Q}_k$ for all $g \in \mathcal{N}$ and all $k \in \mathcal{K}$). Then, it is weakly optimal to assign the asset to a party with inalienable means for expropriation, if one exists. That is, if $\mathcal{N} \backslash \mathcal{E} \neq \varnothing$, then for any $g \in \mathcal{E}$ and $h \in \mathcal{N} \backslash \mathcal{E}$, $TS(h) \geqslant TS(g)$.*

Proof. Fix $g \in \mathcal{E}$ and $h \in \mathcal{N} \backslash \mathcal{E}$. By definition of safety thresholds,

$$\tau_{ij}^{\text{safe}}(g) = \begin{cases} \tau_{gj}^{\star} & \text{if } i = g, \\ 1 & \text{if } i \in \mathcal{E} \backslash \{g\}, \\ \tau_{ij}^{\star} & \text{if } i \notin \mathcal{E}, \end{cases} \qquad \tau_{ij}^{\text{safe}}(h) = \begin{cases} 1 & \text{if } i \in \mathcal{E}, \\ \tau_{ij}^{\star} & \text{if } i \notin \mathcal{E}. \end{cases}$$

Thus, $\tau_{ij}^{\text{safe}}(h) \geqslant \tau_{ij}^{\text{safe}}(g)$ for all i, j. Condition $\mathcal{Q}_k(g) = \mathcal{Q}_k(h)$ for all $k \in \mathcal{K}$ implies $TS(h) \geqslant TS(g)$. □

Corollary 2 states that when ownership does not change the set of active participants, it is optimal to allocate the asset to a party that possesses inalienable means of expropriation (i.e., a party that can expropriate regardless of ownership). Since this party is dangerous irrespective of whether it owns the asset, giving it the asset does not create a new expropriation risk. Conversely, giving the asset to a party with alienable expropriation means would turn a previously safe party into a dangerous one.

4 Applications

Our framework highlights that asset ownership may protect knowledge through either control over production or control over expropriation. For example, Apple chose to design the iPad

chip in-house rather than partner with Intel because it "just didn't want to teach them everything, which they could go and sell to our competitors" (Steve Jobs, quoted in Isaacson, 2011, p. 493). Here, ownership served to control production and limit knowledge disclosure. In contrast, Apple's ownership choices in its China operations are better interpreted as control over expropriation. Specifically, Apple actively shared knowledge with its Chinese suppliers. Although it did not own most production facilities, it owned key production machines, which limited suppliers' ability to expropriate Apple's know-how (McGee, 2025).

We also show ownership has costs: non-owners may not be willing to fully disclose their information to the owner. For example, China's quid pro quo policy requires multinationals to form joint ventures with local partners to access the Chinese market. This structure gives the local partner control over some manufacturing decisions, forcing the multinational to disclose production-relevant information. One implication of this policy is that it may discourage automakers from bringing their most advanced technology to China due to concerns about expropriation (Bai et al., 2025).

In this section, we illustrate how our results translate into predictions about optimal ownership and information sharing in several industry environments. In all applications, there are two downstream parties, 1 and 2, who possess payoff-relevant information and compete in the sense that they are tempted to expropriate each other's knowledge. A third upstream party, 3, can collaborate with them and is comparatively less tempted to engage in expropriation. Formally, suppose that parties' expropriation incentives are symmetric across targets and let $\tau_1^\star \equiv \tau_{12}^\star$, $\tau_2^\star \equiv \tau_{21}^\star$, $\tau_3^\star \equiv \tau_{31}^\star = \tau_{32}^\star$ denote their safety thresholds. Party 3 is less tempted to engage in expropriation:

$$\tau_3^\star > \max\{\tau_1^\star, \tau_2^\star\}. \tag{4.1}$$

Section 4.1 shows how the upstream party can act as a *knowledge hub*. Two competing downstream parties can benefit from pooling knowledge, but they are too tempted to expropriate each other's secrets. A third party that contributes neither private information nor human capital may nonetheless optimally hold the asset solely because it can safely aggregate the rivals' information. We use this case to rationalize knowledge sharing through third-party suppliers and solution providers, as commonly observed in the auto industry (Küpper et al., 2020).

Section 4.2 studies a setting in which downstream firms do not need to share knowledge with each other, but each project requires an inalienable input from the upstream party. We show why firms are reluctant to share information with a competitor-owned subsidiary, and why a vertically integrated firm may optimally spin off its supplier to unlock value from collaboration with rivals. This logic helps explain the *vertical spin-off* of Delphi from General

Motors and provides a rationale for separating Intel's foundry business from its chip-design operations (Fitch, 2025).

Section 4.3 modifies the spin-off setting so that one downstream firm's project does not require the asset owner's active participation. In that case, vertical integration with an informational *Chinese Wall* becomes feasible and optimal: the subsidiary can learn a rival's secrets without those secrets reaching the parent. Comparing this case to Section 4.2 yields a simple technological condition distinguishing when Chinese Walls are feasible from when they are impossible, requiring a spin-off to maximize surplus.

4.1 Knowledge Hubs

Consider an extension of Example 1 involving three parties, $\mathcal{N} = \{1, 2, 3\}$, and a single project, $\mathcal{K} = \{1\}$. The project requires the asset owner to match the states of parties 1 and 2 to succeed, so $\mathcal{R}_1 = \{1, 2\}$. No inalienable actions are required, $\bar{\mathcal{R}}_1 = \varnothing$, and all parties possess the inalienable means to expropriate, so $\mathcal{E} = \varnothing$.

In this environment, only the owner's alienable cooperative actions are payoff-relevant. The planner can therefore keep the two non-owners uninformed, eliminating their expropriation incentives. The remaining constraint is the owner's own temptation: to implement the project, the owner must hold enough information about θ_1 and θ_2 to match them, but not so much as to trigger expropriation. Proposition 1 then implies that the implementable total surplus under owner g is proportional to $\tau_g^\star$, and the optimal owner is the least tempted party: $g^* \in \arg\max_{g \in \mathcal{N}} \tau_g^\star$, so that $g^* = 3$ under (4.1).

The distinctive feature of this example is that the optimal owner (party 3) contributes neither private information nor inalienable inputs to production: θ_3 is payoff-irrelevant and $\bar{\mathcal{R}}_1 = \varnothing$. Party 3 is valuable solely because it has a weaker temptation to expropriate. In this sense, it functions as a *knowledge hub*, enabling greater information pooling among competing firms.

4.2 Optimal Spin-offs

Suppose there are three parties, $\mathcal{N} = \{1, 2, 3\}$, and two projects, $\mathcal{K} = \{1, 2\}$. We interpret parties 1 and 2 as downstream competitors and party 3 as a common upstream supplier. Project $k \in \{1, 2\}$ requires two inputs: an alienable decision by the asset owner and an inalienable input by the supplier. Formally, $\mathcal{R}_1 = \{1\}$, $\bar{\mathcal{R}}_1 = \{(3, 1)\}$, and $\mathcal{R}_2 = \{2\}$, $\bar{\mathcal{R}}_2 = \{(3, 2)\}$. Thus, for project k, the asset owner must match the downstream party k's state, and the supplier must match the same state using its specific human capital.

Expropriation incentives differ by party. The supplier requires the asset to expropriate (alienable expropriation), while the downstream firms can expropriate using inalienable means:

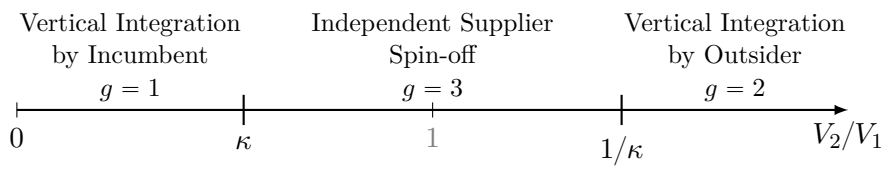

Figure 3: Optimal Organizational Form

The optimal owner varies with the relative market size V_2/V_1. For this figure, we assume that $\kappa^\star < 1$.

$\mathcal{E} = \{3\}$ and $\mathcal{N} \backslash \mathcal{E} = \{1, 2\}$.

We map ownership to organizational form. If a downstream party $i \in \{1, 2\}$ owns the asset, then the supplier is a subsidiary (vertical integration). If the supplier owns the asset ($g = 3$), it is an independent upstream firm.

Given an asset owner $g \in \{1, 2, 3\}$, the cooperative surplus is

$$V_1 \cdot \mathbf{1}\{x_{g1} = \theta_1\} \cdot \mathbf{1}\{\bar{x}_{31} = \theta_1\} + V_2 \cdot \mathbf{1}\{x_{g2} = \theta_2\} \cdot \mathbf{1}\{\bar{x}_{32} = \theta_2\}.$$

Because expropriation destroys surplus, the optimal information policy avoids expropriation and trades it off against the probability of successful cooperation. The key difference across ownership structures is *who must learn a rival's information.* If party 1 owns the asset, then project 1 can be implemented with certainty: party 1 knows θ_1 and the supplier is disarmed (so θ_1 can be safely disclosed to party 3 to satisfy $\bar{x}_{31} = \theta_1$). Project 2 is different. To make the payoff-relevant alienable decision for project 2, the owner must respond to θ_2. Since party 1 is armed and has incentives to expropriate party 2's knowledge, information about θ_2 must be capped at party 1's safety threshold $\tau_1^\star$.

The same logic applies symmetrically if party 2 owns the asset. If instead the supplier owns the asset, then the supplier is armed and must receive (capped) information about both downstream parties' states, so both projects succeed with probability $\tau_3^\star$.

Consequently, the maximal attainable surplus under each ownership assignment is

$$TS(g = 1) = V_1 + \tau_1^\star V_2, \qquad TS(g = 2) = \tau_2^\star V_1 + V_2, \qquad TS(g = 3) = \tau_3^\star (V_1 + V_2).$$

To sharpen the comparative statics, suppose the downstream parties are symmetric in temptation, so $\tau_1^\star = \tau_2^\star \equiv \tau_D^\star$. Recall that the supplier is more trustworthy than a direct competitor, $\tau_3^\star > \tau_D^\star$. Define the threshold

$$\kappa \equiv \frac{1 - \tau_3^\star}{\tau_3^\star - \tau_D^\star}.$$

and assume that $\kappa < 1$. Then vertical integration by the incumbent ($g = 1$) is optimal when the outside market is not too large, $V_2/V_1 < \kappa$. When the outside market is very large, $V_2/V_1 > 1/\kappa$, vertical integration by the outsider ($g = 2$) is optimal. Finally, when the outside market is of intermediate size, $V_2/V_1 \in [\kappa, 1/\kappa]$, the independent supplier (a spin-off, $g = 3$) is optimal.[3]

Figure 3 summarizes the resulting pattern. When the outside market is small, it is optimal for the incumbent to integrate the supplier, because doing so disarms the supplier and lets the incumbent's own project succeed with certainty. As the outside market grows, the opportunity cost of throttling the outsider's project becomes large. At that point, shifting ownership to the (more trustworthy) supplier relaxes the binding information constraint and makes both projects feasible at the higher probability $\tau_3^\star$. Finally, when the outside market is very large, the logic flips and vertical integration by the outsider becomes optimal.

4.3 Chinese Walls Within Organizations

We now modify the setting of Section 4.2 in one important way. Project 1 remains unchanged, but Project 2 no longer requires an alienable action by the asset owner. Formally, $\mathcal{R}_1 = \{1\}$, $\bar{\mathcal{R}}_1 = \{(3, 1)\}$, and $\mathcal{R}_2 = \varnothing$, $\bar{\mathcal{R}}_2 = \{(3, 2)\}$. Project 2 now requires *only* the supplier's inalienable input; the owner does not need to match θ_2.

This change alters the feasible information flows. If party 1 owns the asset ($g = 1$), the supplier is disarmed (because $\mathcal{E} = \{3\}$ and $3 \neq g$). The supplier can therefore be fully informed about both θ_1 and θ_2 and can supply inalienable inputs for both projects. Crucially, because the owner is not an active participant in project 2 (the project has no alienable requirements), party 1 does not need to learn θ_2. The organization can therefore sustain a perfect informational partition: the subsidiary (party 3) learns the rival's information to serve it, while the parent (party 1) remains ignorant, eliminating the expropriation risk.

The resulting attainable surplus levels are

$$TS(g = 1) = V_1 + V_2, \qquad TS(g = 2) = \tau_2^\star V_1 + V_2, \qquad TS(g = 3) = \tau_3^\star (V_1 + V_2).$$

Thus, when $g = 1$, the first-best outcome is achieved.

Comparing this case to Section 4.2 clarifies when Chinese Walls are credible. They are technologically impossible when the owner must be an active participant in the rival's project (because then the owner must be informed about the rival's state). They become credible precisely when the owner can remain operationally passive in the rival's project, so the rival's information can be routed to a disarmed subsidiary without reaching the armed parent.

We note that establishing the theoretical possibility of a Chinese Wall is distinct from the

[3] If $\kappa > 1$, the the optimal ownership is $g^* = 1$ if $V_2/V_1 \leqslant 1$ and $g^* = 2$ if $V_2/V_1 > 1$.

question of how to build it. Questions regarding the specific internal governance mechanisms and incentive systems required to enforce separation between a parent and its subsidiary lie beyond the scope of this framework and represent an exciting avenue for future research.

5 Discussion

This section clarifies the mechanisms driving our main results by analyzing three variations of the baseline model. First, we introduce partial contractibility of actions to show that asset ownership matters for knowledge sharing strictly due to the *non-contractibility of expropriation.* If these actions were contractible, the parties could achieve the first-best regardless of the ownership structure. Second, we show that reallocating ownership often fails to restore the first-best because it confers *a bundle of inextricable* decision rights. Specifically, ownership bundles multiple productive tasks together (forcing a single owner to rely on others' private information) and bundles control over production with the means to expropriate. Unbundling these rights — either by assigning each productive task to the informed party, or by decoupling expropriation capabilities from productive actions — would weakly increase joint surplus and could achieve the first-best. Finally, we relax the assumption of prohibitive expropriation costs to study the mediator's problem under arbitrary costs. We show that the mediator may optimally tolerate some expropriation to achieve a higher probability of cooperative success. Consequently, the optimal owner may be the party whose expropriation imposes the lowest social cost. We show that the information-design problem is closely connected to optimal transport theory.

5.1 Contracts vs. Ownership

In the baseline model, we assume that all actions are non-contractible. If some actions are contractible, then whenever the mediator recommends those actions they are enforced, which relaxes the obedience constraints. Thus, contractibility weakly increases the maximum attainable total surplus. This section outlines the results; formal proofs are provided in Appendix A.3.

If *expropriation* actions $(z, \bar{z})$ are contractible, then the mediator can directly enforce $z_{ij} = \bar{z}_{ij} = 0$ before any information is disclosed. Once expropriation is ruled out, the mediator can fully reveal information and achieve the first-best under *any* ownership structure.

If *cooperative* actions $(x, \bar{x})$ and project shares (α_{ik}) are contractible while expropriation remains non-contractible, then the fundamental constraint is unchanged: after learning information, an armed party can still deviate toward expropriation. Therefore, information disclosure must still satisfy the same *expropriation obedience* (safety-threshold) constraints as

in the baseline model. As a result, the maximal attainable total surplus is governed by the same obedience-based bound as in the baseline problem (i.e., it cannot exceed $\max_g TS(g)$).

If only the cooperative shares (α_{ik}) are contractible (without making $(x, \bar{x})$ contractible), this cannot increase attainable total surplus either. The "shares-only" environment is a restriction of the "contractible cooperation and shares" environment, so its value is weakly smaller. Since the latter yields no improvement over the baseline and the baseline policy remains feasible when shares are contractible, all three values coincide.

5.2 Multiple Assets and Separation of Control Rights

In the baseline model, a single asset bundles multiple alienable decision rights. This bundling generates the central trade-off: ownership protects the owner's knowledge but leaves others exposed to expropriation, discouraging disclosure. This subsection extends the model to $M \geqslant 1$ assets and allows control rights to be unbundled.

Assets. Let $\mathcal{M} = \{1, \ldots, M\}$ be the set of assets. Two assignment functions determine which asset governs which *alienable* decision:

$$\phi^c : \mathcal{N} \to \mathcal{M}, \qquad \phi^e : \mathcal{N} \to \mathcal{M}.$$

Here $\phi^c(j)$ is the asset whose owner controls the alienable *cooperative* action targeting state θ_j, while $\phi^e(j)$ is the asset whose owner controls the alienable *expropriation* action targeting θ_j (for asset-dependent expropriators). The baseline model corresponds to $m = 1$ and $\phi^c = \phi^e \equiv 1$. In general, control over production decisions is bundled when $\phi^c(i) = \phi^c(j)$ for $i \neq j$; control over production and expropriation is bundled when $\phi^c(j) = \phi^e(j)$.

An ownership profile is $g = (g_i)_{i \in \mathcal{M}} \in \mathcal{N}^{\mathcal{M}}$. For each target j, define the induced owners

$$g^c(j) \equiv g_{\phi^c(j)}, \qquad g^e(j) \equiv g_{\phi^e(j)}.$$

Production. Project k succeeds if and only if the relevant alienable and inalienable matching requirements are satisfied:

$$x_{g^c(j)\,j} = \theta_j \quad \forall j \in \mathcal{R}_k, \qquad \bar{x}_{\ell j} = \theta_j \quad \forall (\ell, j) \in \bar{\mathcal{R}}_k.$$

The set of matching tasks for project k under ownership profile g is

$$\mathcal{Q}_k(g) \equiv \{(g^c(j), j) : j \in \mathcal{R}_k\} \ \cup \ \bar{\mathcal{R}}_k. \tag{5.1}$$

Expropriation. A party i can expropriate target $j \neq i$ either via inalienable means ($i \notin \mathcal{E}$) or via an asset if $i \in \mathcal{E}$ and $i = g^e(j)$. Thus the set of parties who are armed *against target* j is

$$\mathrm{Arm}_j(g) \equiv \{i \in \mathcal{N} : i = g^e(j) \text{ or } i \notin \mathcal{E}\}.$$

Define the target-specific safety threshold (and set $\tau_{ii}^{\mathrm{safe}}(g) \equiv 1$)

$$\tau_{ij}^{\mathrm{safe}}(g) \;\equiv\; \begin{cases} \tau_{ij}^{\star} = \dfrac{c_{ij}}{b_{ij} + c_{ij}} & \text{if } i \in \mathrm{Arm}_j(g) \text{ and } i \neq j, \\ 1 & \text{otherwise.} \end{cases} \tag{5.2}$$

Total Surplus. Under the same parameter restrictions as in Proposition 1 (Assumption 1), the characterization of the maximum attainable total surplus carries over:

$$TS(g) = \sum_{k \in \mathcal{K}} V_k \cdot \min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\mathrm{safe}}(g),$$

where $\mathcal{Q}_k(g)$ and $\tau_{ij}^{\mathrm{safe}}(g)$ are defined in (5.1) and (5.2), respectively.

Since the baseline model corresponds to $M = 1$ and $\phi^c = \phi^e \equiv 1$, allowing separable control rights weakly increases maximum attainable total surplus. The following proposition shows how unbundling control rights across assets can achieve the first best.

Proposition 2 (First best under separable control rights)**.** *The first best V_{total} is achievable in each of the following cases.*

(i) All cooperative requirements are alienable ($\bar{\mathcal{R}}_k = \varnothing$ for all k), and distinct targets are controlled by distinct cooperative assets (ϕ^c is injective).

(ii) All expropriation is asset-dependent ($\mathcal{E} = \mathcal{N}$), distinct targets are governed by distinct expropriation assets (ϕ^e is injective).

Proof. (i) Because ϕ^c is injective, we can choose an ownership profile g such that $g_{\phi^c(j)} = j$ for all $j \in \mathcal{N}$, i.e. $g^c(j) = j$ for all j. Since $\bar{\mathcal{R}}_k = \varnothing$, each project has task set $\mathcal{Q}_k(g) = \{(j,j) : j \in \mathcal{R}_k\}$, which is trivial because party j observes θ_j. Hence $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\mathrm{safe}}(g) = 1$ for every k and therefore $TS(g) = \sum_k V_k = V_{\mathrm{total}}$.

(ii) Because ϕ^e is injective, choose g such that $g_{\phi^e(j)} = j$ for all j, i.e. $g^e(j) = j$. Fix any $i \neq j$. Since $\mathcal{E} = \mathcal{N}$, party i has no inalienable expropriation; and because $g^e(j) = j \neq i$, party i also does not control the expropriation asset for target j. Hence $i \notin \mathrm{Arm}_j(g)$ and $\tau_{ij}^{\mathrm{safe}}(g) = 1$ for all $i \neq j$. It follows that for every project k, $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\mathrm{safe}}(g) = 1$, so again $TS(g) = V_{\mathrm{total}}$. □

The intuition behind Proposition 2 is as follows. When cooperation requires only alienable actions and ownership can be arranged so that each action that depends on party i's

information is controlled by party i, then no private information needs to be disclosed. Parties can therefore avoid expropriation because they can remain silent while still implementing the required cooperative decisions. This case extends the mechanism in Example 3 of Section 3.2 (ownership as control over production).

When expropriation requires access to assets and ownership can be arranged so that each party controls the asset needed to expropriate *its own* information, then parties can block all expropriation attempts. In that case, the first best is achievable because the means of expropriation are held by the potential victim rather than the potential expropriator. This case extends the mechanism in Example 4 of Section 3.2 (ownership as control over expropriation).

5.3 Optimal Decision Rule Under Arbitrary Expropriation Costs

In the baseline model, we assume successful expropriation is prohibitively costly, $\lambda_{ij} - b_{ij} > V_{\text{total}}$, so the mediator always prefers to deter expropriation. Here we provide intuition for the general results under arbitrary expropriation costs; Appendix A.4 provides formal details.

When expropriation is costly but not prohibitive, the mediator faces the following tradeoff. Information is "free" up to a party's safety threshold, but pushing accuracy beyond that point requires making expropriation incentive compatible, which entails expected expropriation costs. At the same time, production is O-ring: accuracy for one task is wasted in states where some other required tasks fail. Thus, the mediator wants to "buy" accuracy only when it can be coordinated into joint success.

The general solution admits a two-step characterization: (i) *within-party design*, and (ii) *across-party coupling*. The first step determines for each party i how informative i's private recommendations are across targets (subject to expropriation obedience). This step pins down what i can achieve *on its own* — in particular, the *marginal* distribution of i's success across projects. In the second step, given the induced marginals from Step (i), the mediator chooses how to correlate different parties' project successes to maximize joint (O-ring) output.

We explain these steps in the reverse order: conditional on Step (i), Step (ii) is an optimal-transport problem with no incentive constraints. Because recommendations are private, correlating parties' latent randomization affects outcomes but does not change any party's posterior or incentives. Step (ii) can be interpreted as moving probability mass toward states of *simultaneous* success, and Step (i) can then be interpreted as choosing the "supply" and "demand" that feed into that transport problem.

To build intuition, consider a case with two parties $i \in \{1, 2\}$ and two projects $k \in \mathcal{K} = \{1, 2\}$. Assume that both parties are required for both projects. The general case with arbitrary number of projects, parties, and matching requirements is described in Appendix A.4.

Step (ii): Across-Party Coupling. Any obedient decision rule induces, for each party $i \in \{1,2\}$, a random *success type* $T_i \subseteq \mathcal{K} = \{1,2\}$, where $k \in T_i$ means that party i successfully completes *all* tasks required of i for project k, while $k \notin T_i$ means that i fails in at least one of the tasks in project k. Thus, T_i can take four values: $T_i \in \{\varnothing, \{1\}, \{2\}, \{1,2\}\}$. For notation simplicity, denote $00 \equiv \varnothing$, $10 \equiv \{1\}$, $01 \equiv \{2\}$, and $11 \equiv \{1,2\}$.

Let the *marginal* distribution of T_i be $\beta_i(h) \equiv \mathbb{P}(T_i = h)$ for $h \in \{00, 10, 01, 11\}$. Let the *joint* distribution (a coupling) be $\gamma(h_1, h_2) \equiv \mathbb{P}(T_1 = h_1, T_2 = h_2)$, which can be represented as a 4×4 matrix, so the row sums of γ are β_1 and the column sums are β_2.

Conditional on (h_1, h_2), project k succeeds if and only if $k \in h_1 \cap h_2$. Hence, the cooperative payoff from (h_1, h_2) is $R(h_1, h_2) \equiv V_1 \cdot \mathbf{1}\{1 \in h_1 \cap h_2\} + V_2 \cdot \mathbf{1}\{2 \in h_1 \cap h_2\}$, which can be represented as the following matrix:

$R(h_1, h_2) \equiv$

$h_1 \backslash h_2$	00	10	01	11
00	0	0	0	0
10	0	V_1	0	V_1
01	0	0	V_2	V_2
11	0	V_1	V_2	$V_1 + V_2$

.

Fix the marginals (β_1, β_2). The mediator's best possible cooperative surplus is obtained by choosing γ to solve

$$\max_{\gamma \geqslant 0} \sum_{h_1, h_2} \gamma(h_1, h_2)\, R(h_1, h_2) \quad \text{s.t.} \quad \sum_{h_2} \gamma(h_1, h_2) = \beta_1(h_1), \quad \sum_{h_1} \gamma(h_1, h_2) = \beta_2(h_2),$$

which is a finite Kantorovich optimal transport problem: β_1 is "supply," β_2 is "demand," and γ is the "shipping plan." Because $R(11, 11) = V_1 + V_2$ is the largest entry, the transport optimizer tries to match "strong" types with "strong" types (positive assortative matching), moving mass toward the diagonal (especially $(11, 11)$), subject to the row and column sums. Intuitively, Step (ii) chooses *correlation*: holding fixed what each party can achieve on its own (the marginals), the mediator aligns success profiles to maximize simultaneous project success.

Step (i): Within-Party Design. In our setting, the marginals (β_1, β_2) are not exogenous: they are chosen via information design and the expropriation tradeoff. At a high level, the mediator chooses how often party i is recommended to attempt expropriation and how accurate i's recommendations are conditional on that expropriation pattern. Obedience generates a "free" region and a "paid" region: if i is not recommended to expropriate a target j, posterior accuracy about j cannot exceed the safety threshold $\tau_{ij}^{\text{safe}}(g)$; if i is recommended to attempt

expropriation, accuracy must be between $\tau_{ij}^{\text{safe}}(g)$ and 1. Thus, increasing accuracy shifts β_i away from 00 and toward 10, 01, and 11, but doing so beyond safety thresholds requires incurring expected expropriation costs.

Optimal Decision Rule. To sum up, in Step (i), the mediator chooses the distribution of expropriation attempt patterns (which targets each party attempts to expropriate and how often) and the accuracy of recommendations (the probability that they are correct) conditional on an attempt pattern. In Step (ii), the mediator chooses the correlation of success across parties (γ) conditional on marginal success (β_i) determined in Step (i).

In optimal transport language, Step (i) chooses the "supply" and "demand" by reshaping each β_i (incurring expropriation costs when necessary), and Step (ii) then solves the transport problem with the payoff matrix R to align these marginals across parties. For arbitrary number of projects and parties, Step (ii) is a multi-marginal optimal transport problem.

When expropriation is sufficiently costly (the baseline model), it is optimally deterred, so accuracies are capped at safety thresholds. Then, in an optimal decision rule, each party's feasible success profiles lie on a *ladder*: in the two-project case, the support of T_i is either $\{00, 01, 11\}$ or $\{00, 10, 11\}$ (one of 10 and 01 has zero probability). This ranking simplifies Step (ii), as the optimal coupling γ collapses to a one-dimensional comonotone construction: there exists a common $\xi \sim \text{Unif}[0, 1]$ such that each T_i is a deterministic function of ξ. Visually, the support of the optimal coupling traces a one-dimensional monotone path through matrix R. This insight generalizes to cases with many projects and parties.

With arbitrary expropriation costs, the mediator may optimally randomize over expropriation attempt patterns. Conditional on a fixed pattern (i.e., the set of targets for which party i is recommended expropriation), it is still optimal to make task-level correctness comonotone within the party, so *conditional* T_i lies on a ladder. However, different expropriation patterns can induce different ladder orderings across projects, and mixing across patterns can therefore generate crossing types (both 10 and 01 with positive probability).

A simple sufficient condition that rules out this mixing problem is an ordered (nested) production technology: if projects can be indexed so that $\mathcal{Q}_1(g) \supseteq \mathcal{Q}_2(g) \supseteq \cdots \supseteq \mathcal{Q}_{|\mathcal{K}|}(g)$ (Project 1 is the "hardest", and Project $|\mathcal{K}|$ is the "easiest"), then the induced ordering of project success is the same across all attempt patterns, so the unconditional T_i remains ladder-valued. In this case, the optimal coupling γ again collapses to a one-dimensional comonotone (single-uniform) construction; without such monotonicity, crossing types can be optimal and the full multi-marginal transport problem is required.

Optimal Ownership. Up to this point, we have described how to find the optimal decision rule for a given ownership structure $g \in \mathcal{N}$. Once this rule is characterized, the optimal owner

is the party that maximizes total surplus. A key difference relative to the baseline model is that this assignment may now depend on the absolute social cost of expropriation.

To build intuition, consider the extension of Example 3 (where an owner requires a non-owner's knowledge for production) under arbitrary expropriation costs: two parties $\mathcal{N} = \{1, 2\}$ and a single project $\mathcal{K} = \{1\}$. The project requires only alienable actions ($\mathcal{R}_1 = \{1, 2\}$ and $\bar{\mathcal{R}}_1 = \varnothing$), and each party has the ability to expropriate ($\mathcal{E} = \varnothing$).

Since the non-owner does not participate in production, only the owner must be informed about the non-owner's information. This problem can be analyzed using Bayesian Persuasion techniques as in Example 1. Applying the logic of Figure 1, the optimal decision rule takes one of two forms: either partial disclosure up to a safety threshold (optimal when expropriation costs are high) or full disclosure (optimal when expropriation costs are low). Thus, the maximum total surplus under an owner g is

$$TS(g) = \max\{\tau_g^\star \cdot V_1, V_1 - (\lambda_g - b_g)\}$$

In the baseline model, the social cost of expropriation $\lambda_g - b_g$ is high for both $g \in \{1, 2\}$. Consequently, the second term is always smaller, and the optimal owner is the party with the lower temptation to expropriate (higher $\tau_g^\star$).

Suppose now that party 1 has a higher temptation to expropriate ($\tau_1^\star < \tau_2^\star$), but its expropriation is "cheaper" ($\lambda_1 - b_1 < V_1 < \lambda_2 - b_2$). If $\lambda_1 - b_1$ is sufficiently close to zero, it becomes optimal to assign the asset to party 1 — enabling full efficiency at a small cost — even though it has a higher temptation to expropriate.

6 Conclusion

This paper highlights that when knowledge can be used both productively and opportunistically, the central problem is coordinating its allocation across parties. In the absence of contractual tools to prevent opportunistic behavior, ownership mediates this friction by jointly shaping production requirements and expropriation capabilities. However, reallocating ownership is an imperfect substitute for complete contracts: such transfers inherently move a coarse bundle of decision rights, rather than providing the targeted control necessary to achieve efficient cooperation without the risk of expropriation.

This framework suggests a few directions for future work. First, our model takes a step toward studying how transactions across firms are shaped by their internal structures. In particular, we provide a simple condition on the production technology that makes internal "Chinese Walls" within organizations impossible. However, even if such walls are theoretically

possible, their credibility still depends on governance and internal incentives. Incorporating richer organizational structure into this framework is a natural next step.

Second, in our mechanism-design formulation, parties commit to a mediated contract before observing their private information. If contracting is instead initiated by an informed party, the contract offer itself might convey information. In standard economic models, private information is usually represented as the realization of a random variable observed by some parties but not others, drawn from a common-knowledge distribution. In such settings, information can in principle be concealed through the design of the mechanism — an insight known as the *inscrutability principle* (Myerson, 1983).

The inscrutability logic can fail when information is "eye-opening," in the sense that it changes how parties perceive the relevant state space or feasible actions (Tirole, 2009). Then the mere act of specifying previously unforeseen contingencies expands the other party's awareness and shifts beliefs. Even if parties can contractually rule out expropriation, the disclosure required to initiate contracting changes outside options relative to no contracting, so the initial ownership structure can shape knowledge flows even when expropriation is contractible. This is close in spirit to Grossman and Hart (1986), where making an offer functions like an investment that shifts outside options — except here the relevant object is *initial* rather than final ownership. Formalizing this setting likely requires an alternative model of information (e.g., Akerlof, Holden, and Li, 2025) and is a promising direction for future research.

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A Appendix

A.1 Proof of Proposition 1

Recall that $\mathcal{R}_k$ and $\bar{\mathcal{R}}_k$ are alienable and inalienable requirements, respectively, and $\mathcal{Q}_k(g) \equiv \{(g,j) : j \in \mathcal{R}_k\} \cup \bar{\mathcal{R}}_k \subseteq \mathcal{N} \times \mathcal{N}$ is the set of production requirements under owner $g \in \mathcal{N}$. Proposition 1 states that the maximum attainable expected total surplus under owner $g \in \mathcal{N}$ is

$$TS(g) = \sum_{k \in \mathcal{K}} V_k \cdot \left(\min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) \right). \tag{A.1}$$

Moreover, there exists an optimal obedient decision rule that is *nested*. Let $\xi \sim \text{Unif}[0,1]$ be a common randomization device. For each (ξ, θ, g) define an action profile $a(\xi, \theta, g) = (x, \bar{x}, z, \bar{z})$ as follows:

$$z_{ij} = 0 \;\text{ and }\; \bar{z}_{ij} = 0 \qquad \text{for all } i \in \mathcal{N},\; j \neq i, \tag{A.2}$$

$$x_{ij} = \bar{x}_{ij} = \begin{cases} \theta_j & \text{if } \xi \leqslant \tau_{ij}^{\text{safe}}(g), \\ -\theta_j & \text{if } \xi > \tau_{ij}^{\text{safe}}(g), \end{cases} \qquad \text{for all } i \in \mathcal{N}. \tag{A.3}$$

Define $\sigma^g(\cdot \mid \theta) \in \Delta(\mathcal{A})$ as the distribution induced by $\xi \sim \text{Unif}[0,1]$ and $a = a(\xi, \theta; g)$; that is,

$$\sigma^g(a \mid \theta) = \mathbb{P}_\xi\big[a(\xi, \theta; g) = a\big].$$

We prove the proposition in two steps. Lemma 1 shows that this decision rule is obedient and achieves $TS(g)$. Lemma 2 shows that for any obedient decision rule, the expected total surplus does not exceed $TS(g)$.

Lemma 1 (Nested rule is obedient and achieves $TS(g)$)**.** *The decision rule σ^g is obedient under ownership $g \in \mathcal{N}$. Moreover, each project k succeeds with probability $\min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g)$, and hence*

$$\mathbb{E}_\psi\Big[TS(a, g; \theta)\Big] = \sum_{k \in \mathcal{K}} V_k \cdot \min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g) = TS(g),$$

where $\psi(\theta, a) = p(\theta)\sigma^g(a \mid \theta)$ is the induced distribution over actions and states.

Proof. We verify obedience and compute the induced success probabilities.

Posteriors. Party i observes (θ_i, a_i) before choosing actions. The expropriation recommendations in (A.2) are constant and uninformative. Since $x_{ij} = \bar{x}_{ij}$ for all i, j in (A.3), for notational convenience we will keep track of the vector $x_i \equiv (x_{i1}, \ldots, x_{iN})$ only.

Fix i and write $x_{i,-i} \equiv (x_{ij})_{j \neq i}$. For each $j \neq i$, let $s_{ij}(\xi) \in \{-1, 1\}$ be a deterministic sign defined by $s_{ij}(\xi) = 1$ if $\xi \leqslant \tau_{ij}^{\text{safe}}(g)$ and $s_{ij}(\xi) = -1$ otherwise, so that $x_{ij} = s_{ij}(\xi) \cdot \theta_j$. any vector $v \in \{-1, 1\}^{N-1}$ and any $\xi \in [0, 1]$, independence and uniformity of $(\theta_j)_{j \neq i}$ imply

$$\begin{aligned} \mathbb{P}_\psi(x_{i,-i} = v \mid \xi) &= \mathbb{P}_\psi\big(s_{ij}(\xi) \cdot \theta_j = v_j \ \forall j \neq i \,\big|\, \xi\big) \\ &= \prod_{j \neq i} \mathbb{P}_\psi\big(\theta_j = s_{ij}(\xi) \cdot v_j\big) = \prod_{j \neq i} \tfrac{1}{2} = 2^{-(N-1)}. \end{aligned} \tag{A.4}$$

This conditional probability does *not* depend on ξ. Hence $x_{i,-i}$ is independent of ξ. Moreover, since θ_i is independent of $(\xi, x_{i,-i})$, ξ is also independent $(\theta_i, x_{i,-i})$.

Equivalently, Bayes' rule yields that for any $v \in \{-1, 1\}^{N-1}$, the conditional density of $\xi \mid x_i$ is

$$f_{\xi|x_i}(\xi \mid v) = \frac{\mathbb{P}_\psi(x_i = v \mid \xi)\, f_\xi(\xi)}{\mathbb{P}_\psi(x_i = v)} = \frac{2^{-(N-1)} \cdot 1}{2^{-(N-1)}} = 1,$$

so $\xi \mid X_i$ is uniform on $[0, 1]$. Since θ_i is independent of (ξ, x_i), we also have $f_{\xi|\theta_i,x_i}(\cdot \mid \theta_i, x_i) \equiv 1$.

Fix $j \neq i$. Under the nested rule, $\theta_j = x_{ij}$ if and only if $\xi \leqslant \tau_{ij}^{\text{safe}}(g)$. Therefore, using iterated expectations and the posterior in (A.4),

$$\begin{aligned} \mathbb{P}_\psi(\theta_j = x_{ij} \mid \theta_i, x_i) &= \mathbb{P}_\psi(\xi \leqslant \tau_{ij}^{\text{safe}}(g) \mid \theta_i, x_i) \\ &= \int_0^{\tau_{ij}^{\text{safe}}(g)} f_{\xi|\theta_i,x_i}(\xi \mid \theta_i, x_i)\, d\xi = \int_0^{\tau_{ij}^{\text{safe}}(g)} 1\, d\xi = \tau_{ij}^{\text{safe}}(g). \end{aligned}$$

No expropriation. The recommendation rule always prescribes no expropriation (A.2). If i is disarmed, then every effective expropriation action is 0 and there is no deviation. If i is armed, then $\tau_{ij}^{\text{safe}}(g) = \tau_{ij}^{\star} = c_{ij}/(b_{ij} + c_{ij})$ by definition, so conditional on (θ_i, a_i) the maximal success probability from attempting expropriation against any target is at most $\tau_{ij}^{\star}$, implying that the expected private gain from expropriation is weakly non-positive. With the tie-breaking rule favoring the safe action, obeying (A.2) is optimal.

Obedient Cooperation. Now consider deviations that change i's cooperative actions $(x_i, \bar{x}_i)$. For any project k, let $J_{i,k}(g)$ denote the set of indices $j \neq i$ such that $(i, j) \in \mathcal{Q}_k(g)$, i.e., the states that i must match (via either x_{ij} if $i = g$ and $j \in \mathcal{R}_k$, or via $\bar{x}_{ij}$ if $(i, j) \in \bar{\mathcal{R}}_k$). If $J_{i,k}(g) = \varnothing$, then i's cooperative action does not affect project k.

Fix a project k where $J_{i,k}(g) \neq \varnothing$. Under the nested rule, if all parties follow their recommendations, project k succeeds if and only if $\xi \leqslant \tau_{ij}^{\text{safe}}(g)$ for all required tasks (i, j), which is equivalent to:

$$\xi \leqslant \min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g). \tag{A.5}$$

Since $\tau_{ij}^{\text{safe}}(g) > 1/2$ for all pairs, this probability is strictly greater than 1/2.

Now suppose party i deviates from its recommendation for a subset of targets $\tilde{J}_{i,k}(g) \subseteq J_{i,k}(g)$. Party i successfully matches the states if and only if the recommendation was correct for the non-deviated targets ($j \notin \tilde{J}_{i,k}(g)$) and incorrect for the deviated targets ($j \in \tilde{J}_{i,k}(g)$). In terms of the randomization variable ξ, this requires:

$$\xi \in \Big(\max_{j \in \tilde{J}_{i,k}(g)} \tau_{ij}^{\text{safe}}(g), \min_{j \in J_{i,k}(g) \setminus \tilde{J}_{i,k}(g)} \tau_{ij}^{\text{safe}}(g) \Big], \quad \text{with the convention } \min \varnothing \equiv 1.$$

Since $\tau_{ij}^{\text{safe}}(g) > 1/2$, the lower bound of this interval is strictly greater than 1/2. Therefore, the length of the interval — and thus the success probability — is strictly less than 1/2. Consequently, any deviation yields a strictly lower probability of success than obedience.

Total surplus. Multiplying the probability of the event in (A.5) by V_k and summing over all projects yields the stated expression for $TS(g)$. □

Lemma 2 (Upper bound for any obedient rule)**.** *Fix ownership $g \in \mathcal{N}$. Let $\sigma(\cdot \mid \theta)$ be any obedient decision rule under g, and let $\psi(\theta, a) = p(\theta)\sigma(a \mid \theta)$ be the induced distribution. Then*

$$\mathbb{E}_{\psi}\left[TS(a, g; \theta)\right] \leqslant TS(g).$$

Proof. For each task $(i, j) \in \mathcal{Q}_k(g)$ define the event that this task is satisfied:

$$C_{k,(i,j)} \equiv \begin{cases} \{(\theta, a) : x_{ij} = \theta_j\} & \text{if } (i,j) = (g,j) \in \{(g,j) : j \in \mathcal{R}_k\} \text{ and } (i,j) \notin \bar{\mathcal{R}}_k, \\ \{(\theta, a) : \bar{x}_{ij} = \theta_j\} & \text{if } (i,j) \in \bar{\mathcal{R}}_k \text{ and } (i,j) \notin \{(g,j) : j \in \mathcal{R}_k\}, \\ \{(\theta, a) : x_{ij} = \bar{x}_{ij} = \theta_j\} & \text{if } (i,j) = (g,j) \in \{(g,j) : j \in \mathcal{R}_k\} \cap \bar{\mathcal{R}}_k. \end{cases} \tag{A.6}$$

Let E_k denote the event that project k succeeds:

$$E_k \equiv \bigcap_{(i,j) \in \mathcal{Q}_k(g)} C_{k,(i,j)}.$$

We write total surplus as

$$TS(a, g; \theta) = TS^{\text{coop}}(a, g; \theta) + TS^{\text{exp}}(a, g; \theta),$$

where $TS^{\text{coop}}(a, g; \theta) = \sum_{k \in \mathcal{K}} V_k \, \mathbf{1}\{E_k\}$, and TS^{exp} is the sum of all expropriation terms.

Upper Bound on Task Probabilities. Let

$$r_{ij} \equiv \mathbb{P}_\psi\left(\zeta_{ij} \neq 0\right)$$

be the probability that party i attempts an expropriation of party j's information, $j \neq i$. If $i \notin \mathrm{Arm}(g)$, then party i cannot engage in expropriation and $r_{ij} = 0$.

Decompose $\mathbb{P}_\psi(C_{k,(i,j)})$ according to whether i attempts to expropriate j:

$$\mathbb{P}_\psi(C_{k,(i,j)}) = \mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} = 0) + \mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} \neq 0)$$

On the event $\{\zeta_{ij} = 0\}$, obedience implies that attempting expropriation is not strictly profitable for party i. Therefore, conditional on (θ_i, a_i), the probability that any sign choice satisfies the matching task is at most $\tau_{ij}^{\mathrm{safe}}(g)$, so

$$\mathbb{P}_\psi\left(C_{k,(i,j)} \mid \zeta_{ij} = 0\right) \leqslant \tau_{ij}^{\mathrm{safe}}(g) \Rightarrow \mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} = 0) \leqslant \tau_{ij}^{\mathrm{safe}}(g)\mathbb{P}_\psi\left(\zeta_{ij} = 0\right) = \tau_{ij}^{\mathrm{safe}}(g)(1 - r_{ij})$$

In addition,

$$\mathbb{P}_\psi(C_{k,(i,j)}, \zeta_{ij} \neq 0) \leqslant \mathbb{P}_\psi\left(\zeta_{ij} \neq 0\right) = r_{ij}$$

Combining these two inequalities, we get

$$\mathbb{P}_\psi(C_{k,(i,j)}) \leqslant \tau_{ij}^{\mathrm{safe}}(g)(1 - r_{ij}) + r_{ij}$$

Upper bound on cooperative surplus. The goal is to show that the probability of project k's success has the following upper bound

$$\mathbb{P}_\psi\left(E_k\right) \leqslant \min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\mathrm{safe}}(g)$$

Fix project $k \in \mathcal{K}$. If $\mathcal{Q}_k(g)$ contains only trivial tasks of the form (i,i), then $\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\mathrm{safe}}(g) = 1$, so the desired bound holds.

Fix any project k with non-trivial tasks: $(i,j) \in \mathcal{Q}_k(g)$ with $i \neq j$. Pick one bottleneck task:

$$(i_k, j_k) \in \arg\min_{(i,j)\in\mathcal{Q}_k(g)} \tau_{ij}^{\mathrm{safe}}(g) \quad \text{with } j_k \neq i_k.$$

Since $E_k \subseteq C_{k,i_k,j_k}$, we have

$$\mathbb{P}_\psi(E_k) \leqslant \tau_{i_k j_k}^{\mathrm{safe}}(g) + \left(1 - \tau_{i_k j_k}^{\mathrm{safe}}(g)\right) r_{i_k j_k},$$

Multiplying by V_k and summing over projects yields

$$\mathbb{E}_\psi[TS^{\text{coop}}] \leqslant TS(g) + \sum_{i\in\text{Arm}(g)} \sum_{j\neq i} \big(1 - \tau_{ij}^{\text{safe}}(g)\big)\, V_{ij}^{\min}\, r_{ij}, \tag{A.7}$$

where

$$V_{ij}^{\min} \equiv \sum_{k:\ (i_k,j_k)=(i,j)} V_k.$$

Upper bound on expropriation surplus. Fix a pair (i,j) with $i \in \text{Arm}(g)$ and $j \neq i$. On the event $\{\zeta_{ij} \neq 0\}$, obedience implies that attempting expropriation is (weakly) optimal, hence the conditional success probability is at least the private threshold:

$$\mathbb{P}_\psi(\zeta_{ij} = \theta_j \mid \zeta_{ij} \neq 0) \ \geqslant\ \tau_{ij}^\star.$$

Therefore the expected welfare contribution from this ordered pair satisfies

$$\mathbb{E}_\psi\Big[(b_{ij} - \lambda_{ij})\, \mathbf{1}\{\zeta_{ij} = \theta_j\} - c_{ij}\, \mathbf{1}\{\zeta_{ij} = -\theta_j\}\Big] \leqslant \mathbb{E}_\psi\Big[(b_{ij} - \lambda_{ij})\, \mathbf{1}\{\zeta_{ij} = \theta_j\}\Big] \leqslant (b_{ij} - \lambda_{ij})\, \tau_{ij}^\star\, r_{ij},$$

where we the first inequality follows from $-c_{ij} < 0$ and the second inequality follows from the fact the probability of successful expropriation should be at least $\tau_{ij}^\star$. Summing over (i,j) gives

$$\mathbb{E}_\psi[TS^{\text{exp}}] \leqslant \sum_{i\in\text{Arm}(g)} \sum_{j\neq i} \tau_{ij}^\star\, (b_{ij} - \lambda_{ij})\, r_{ij}. \tag{A.8}$$

Upper bound on total surplus. Combining (A.7) and (A.8) gives

$$\begin{aligned}
\mathbb{E}_\psi[TS(a,g;\theta)] &\leqslant TS(g) + \sum_{i\in\text{Arm}(g)} \sum_{j\neq i} r_{ij}\Big[(1-\tau_{ij}^\star)V_{ij}^{\min} + \tau_{ij}^\star(b_{ij} - \lambda_{ij})\Big] \\
&= TS(g) - \sum_{i\in\text{Arm}(g)} \sum_{j\neq i} r_{ij}\Big[\underbrace{\tau_{ij}^\star(\lambda_{ij} - b_{ij}) - \big(1-\tau_{ij}^\star\big)V_{ij}^{\min}}_{>0}\Big].
\end{aligned}$$

By Assumption 1, $\lambda_{ij} - b_{ij} > V_{\text{total}} \geqslant V_{ij}^{\min}$. Also $b_{ij} < c_{ij}$ implies $\tau_{ij}^\star = \frac{c_{ij}}{b_{ij}+c_{ij}} > \frac{1}{2}$, hence $\tau_{ij}^\star > (1-\tau_{ij}^\star)$. Therefore, for every $i \in \text{Arm}(g)$,

$$\tau_{ij}^\star(\lambda_{ij} - b_{ij}) \ >\ \tau_{ij}^\star V_{ij}^{\min} \ >\ (1-\tau_{ij}^\star)V_{ij}^{\min},$$

so the bracketed term is strictly positive. Therefore, the maximum attainable expected total surplus is $TS(g)$ and can be attained only when there is no expropriation, $r_{ij} = 0$ for all $i \in \mathcal{N}$. □

Conclusion. Lemma 1 shows $TS(g)$ is attainable by the nested no-expropriation rule. Lemma 2 shows no obedient rule can exceed it under Assumption 1. This completes the proof of Proposition 1.

A.2 Implementation Through a Mediated Contract

Recall that each party $i \in \mathcal{N} = \{1, \ldots, N\}$ privately observes a binary state $\theta_i \in \Theta_i \equiv \{-1, 1\}$. Let $\Theta \equiv \prod_{i \in \mathcal{N}} \Theta_i$ and assume the common prior is independent and uniform: $p(\theta) = \left(\frac{1}{2}\right)^N$.

For each party i, the set of feasible action plans is $\mathcal{A}_i \equiv \{-1, 1\}^{2N} \times \{-1, 0, 1\}^{2(N-1)}$. A non-contractible action plan $a_i \in \mathcal{A}_i$ can be written as $a_i = \big((x_{ij})_{j \in \mathcal{N}}, (\bar{x}_{ij})_{j \in \mathcal{N}}, (z_{ij})_{j \neq i}, (\bar{z}_{ij})_{j \neq i}\big)$, and we write $\mathcal{A} \equiv \prod_{i \in \mathcal{N}} \mathcal{A}_i$ and $a = (a_i)_{i \in \mathcal{N}} \in \mathcal{A}$.

Ownership $g \in \mathcal{N}$ is contractible. Transfers are contractible and budget balanced:

$$t \in \mathcal{T} \equiv \Big\{\tilde{t} \in \mathbb{R}^N : \sum_{i \in \mathcal{N}} \tilde{t}_i = 0\Big\}.$$

Let a non-transfer *outcome* be

$$o = (a, g) \in \mathcal{O} \equiv \mathcal{A} \times \mathcal{N}.$$

Given $o = (a, g)$ and state θ, payoffs are $\tilde{u}_i(o, t; \theta) = \tilde{u}_i(a, g, t; \theta) = u_i(a, g; \theta) + t_i$, and total surplus is $TS(a, g; \theta) \equiv \sum_{i \in \mathcal{N}} u_i(a, g; \theta)$, where transfers cancel out.

Contracts. We model contracting as a mediated *coordination mechanism* in the sense of Myerson (1982, 1983). Production and expropriation actions $(x, \bar{x}, z, \bar{z})$ are non-contractible, but parties can contract on (i) asset ownership $g \in \mathcal{N}$ (control of payoff-relevant alienable actions), (ii) monetary transfers $t \in \mathcal{T}$, and (iii) the behavior of a neutral mediator who can receive messages from the parties and privately communicate recommendations to them.

A *direct mediated contract* is a pair (φ, t) consisting of:

1. an *allocation rule* $\varphi(\cdot \mid \hat{\theta}) \in \Delta(\mathcal{O})$ for each report profile $\hat{\theta} \in \Theta$; and
2. a *transfer rule* $t : \mathcal{O} \times \Theta \to \mathcal{T}$, where $t(o, \hat{\theta})$ is the transfer vector implemented when the mediator draws outcome $o = (a, g)$ after receiving report profile $\hat{\theta}$.

The interpretation is as follows. After θ is realized, each party i reports $\hat{\theta}_i \in \Theta_i$ to the mediator. Given $\hat{\theta}$, the mediator draws $(a, g) \sim \varphi(\cdot \mid \hat{\theta})$, enforces ownership g and transfers $t((a, g), \hat{\theta})$, and privately recommends a_i to party i. Parties then simultaneously choose their (non-contractible) action plans in $\mathcal{A}_i$.

Bargaining and timing. Fix an initial owner $g_0 \in \mathcal{N}$. The timing of the game is the following:

1. Party 1 makes a take-it-or-leave-it offer of a contract to the other parties.
2. Each party $i \in \{2, \ldots, N\}$ either accepts or rejects the contract.
3. State $\theta \in \Theta$ is realized. Party i observes θ_i.
4. If all participates accepted, they send their reports $\hat{\theta}_i \in \Theta_i$ to the mediator. The mediator draws $o \in \mathcal{O}$ according to $\varphi(\cdot \mid \hat{\theta})$, enforces contractible decisions and sends private recommendations for non-contractible decisions. If at least one party rejects, then no mediator is appointed, the asset owner is the initial owner g_0, no transfers are shared, and no information is shared. In this disagreement outcome, all parties optimally choose no expropriation and set $x_{ij} = 1$ and $\bar{x}_{ij} = 1$ for all (i, j).
5. Parties observe enforced ownership and transfers.
6. Parties choose their actions.
7. Payoffs are realized.

Incentive Constraints. Following Myerson (1982, 1983), we restrict attention to direct mechanisms and impose incentive compatibility in the sense that truthful reporting and obedience to recommendations form a Bayesian Nash equilibrium.

Fix a contract (φ, t). When all parties report truthfully and follow their recommendations, party i's interim expected utility is

$$U_i(\varphi, t \mid \theta_i) \;\equiv\; \sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{(a,g) \in \mathcal{O}} \varphi(a, g \mid \theta_i, \theta_{-i}) \Big[u_i(a, g; \theta_i, \theta_{-i}) + t_i\big((a, g), (\theta_i, \theta_{-i})\big) \Big].$$

Parties are not compelled to follow recommendations; a (pure) deviation for party i is a function $\delta_i : D_i \to \mathcal{A}_i$ mapping the available information at the action stage into an action plan, where $D_i \equiv N \times \mathcal{T} \times \mathcal{A}_i \times \Theta_i$, and party i observes $d_i \equiv (g, t, a_i, \theta_i) \in D_i$. For any alternative report $\theta_i' \in \Theta_i$ and any deviation function $\delta_i : D_i \to \mathcal{A}_i$, define the *double-deviation* payoff

$$\begin{aligned} U_i^*(\varphi, t, \delta_i, \theta_i' \mid \theta_i) \;\equiv\; \sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{(a,g) \in \mathcal{O}} \varphi(a, g \mid \theta_i', \theta_{-i}) \Big[& u_i\big((\delta_i(g, t', a_i, \theta_i), a_{-i}), g; \theta_i, \theta_{-i}\big) \\ & + t_i\big((a, g), (\theta_i', \theta_{-i})\big) \Big], \end{aligned}$$

where $t' \equiv t((a, g), (\theta_i', \theta_{-i}))$ is the transfer vector implemented after the report (θ_i', θ_{-i}).

The contract (φ, t) is *incentive compatible* if, for every $i \in \mathcal{N}$,

$$U_i(\varphi, t \mid \theta_i) \;\geqslant\; U_i^*(\varphi, t, \delta_i, \theta_i' \mid \theta_i) \qquad \text{for all } \theta_i, \theta_i' \in \Theta_i \text{ and all } \delta_i : D_i \to \mathcal{A}_i. \tag{A.9}$$

This condition subsumes truth-telling (take $\delta_i(d_i) = a_i$) and obedience (take $\theta_i' = \theta_i$) as special cases.

Participation Constraints. Let $o^0 = (\hat{a}^0, g_0)$ denote the disagreement outcome, where $\hat{a}^0$ recommends $x_{ij} = 1$, $\bar{x}_{ij} = 1$ and $z_{ij} = \bar{z}_{ij} = 0$ for all feasible pairs (i, j). The disagreement transfers are zero for all parties, $t_i^0 = 0$. Let $\underline{U}_i$ denote party i's ex-ante expected payoff in disagreement:

$$\underline{U}_i \equiv \sum_{\theta \in \Theta} p(\theta)\, u_i(a^0, g_0; \theta),$$

where $p(\theta) = \frac{1}{2^n}$. A contract (φ, t) is *individually rational* if

$$\sum_{\theta_i \in \Theta_i} p(\theta_i)\, U_i(\varphi, t \mid \theta_i) \;\geqslant\; \underline{U}_i \qquad \text{for all } i \in \mathcal{N} \backslash \{1\}. \tag{A.10}$$

Party 1 can always guarantee itslef $\underline{U}_1$ by offering the null contract; thus we focus on the constraints (A.10) for $i \neq 1$.

The mechanism-design program. Party 1 chooses a contract (φ, t) to maximize her ex-ante expected utility subject to incentive compatibility and participation:

$$\begin{aligned} \max_{\varphi, t} \quad & \sum_{\theta_1 \in \Theta_1} p(\theta_1) U_1(\varphi, t \mid \theta_1) \\ \text{subject to} \quad & \text{(A.9) and (A.10)}, \\ & \varphi(o \mid \theta) \geqslant 0, \quad \sum_{o \in \mathcal{O}} \varphi(o \mid \theta) = 1 \quad \text{for all } \theta \in \Theta, \\ & \sum_{i \in \mathcal{N}} t_i(a, g, \theta) = 0 \quad \text{for all } (a, g, \theta) \in \mathcal{A} \times \mathcal{N} \times \Theta. \end{aligned} \tag{A.11}$$

Relaxed Problem: Obedience Only

Consider the relaxation of (A.11) obtained by dropping the truth-telling constraints (and hence also the double-deviation constraints). In this relaxation, we evaluate incentives under the *truthful* report profile and impose only obedience of the recommended non-contractible actions, together with individual rationality.

Because transfers are enforced before actions are chosen and cannot depend on the realized (non-contractible) actions, a deviation by party i affects only the non-monetary payoff component. Thus, the obedience constraints are: for every $i \in \mathcal{N}$, every $\theta_i \in \Theta_i$, every $g \in \mathcal{N}$, every recommended action plan $a_i \in \mathcal{A}_i$, and every deviation $a_i' \in \mathcal{A}_i$,

$$\sum_{\theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a_{-i}} \varphi\left(a_i, a_{-i}, g \mid \theta_i, \theta_{-i}\right) \left[u_i\left(a_i, a_{-i}, g; \theta\right) - u_i\left(a_i', a_{-i}, g; \theta\right)\right] \geqslant 0 \tag{A.12}$$

Since transfers are budget balanced and payoffs are quasilinear, we have for any contract

$$\mathbb{E}[\tilde{u}_1] = \mathbb{E}\left[\sum_{i\in\mathcal{N}} u_i(a,g;\theta)\right] - \sum_{i\neq 1}\mathbb{E}[\tilde{u}_i] = \mathbb{E}[TS(a,g;\theta)] - \sum_{i\neq 1}\mathbb{E}[\tilde{u}_i].$$

Under ex-ante participation, $\mathbb{E}[\tilde{u}_i] \geqslant \underline{U}_i$ for all $i \neq 1$, so party 1's ex-ante payoff is bounded by

$$\mathbb{E}[\tilde{u}_1] \;\leqslant\; \mathbb{E}[TS(a,g;\theta)] - \sum_{i\neq 1}\underline{U}_i.$$

Moreover, for any obedient allocation rule φ (hence any induced distribution of (θ, g, a) under obedient play), we can choose *constant* balanced transfers $t \in \mathcal{T}$ so that $\mathbb{E}[\tilde{u}_i] = \underline{U}_i$ for every $i \neq 1$. Indeed, letting $\bar{U}_i \equiv \mathbb{E}[u_i(a,g;\theta)]$ denote party i's expected non-monetary payoff under φ, set

$$t_i = \underline{U}_i - \bar{U}_i \quad (i \neq 1), \qquad t_1 = -\sum_{i\neq 1} t_i.$$

These transfers do not affect obedience (A.12) because the same t_i is added to both sides of any action-deviation comparison.

Therefore, maximizing party 1's expected payoff subject to obedience and ex-ante participation is equivalent (up to the additive constant $-\sum_{i\neq 1}\underline{U}_i$) to maximizing expected total surplus subject only to obedience constraint:

$$\begin{aligned} \max_{\varphi} \quad & \sum_{\theta\in\Theta} p(\theta) \sum_{g\in\mathcal{N}} \sum_{a\in\mathcal{A}} \varphi(a,g \mid \theta)\, TS(a,g;\theta) \\ \text{subject to} \quad & \text{(A.12)}, \qquad \varphi(a,g \mid \theta) \geqslant 0, \quad \sum_{a,g}\varphi(a,g \mid \theta) = 1 \;\text{ for all } \theta \in \Theta. \end{aligned} \tag{A.13}$$

For any *fixed* ownership assignment $g \in \mathcal{N}$, if we restrict attention to allocation rules that put probability one on g (i.e., $\varphi(a, g' \mid \theta) = 0$ for $g' \neq g$), then (A.13) coincides with the solution-concept problem (2.2) defining $TS(g)$.

Lemma 3 (Deterministic ownership)**.** *In the relaxed (obedience-only) problem* (A.13)*, allowing randomization of ownership g does not increase the maximal attainable total surplus. There exists an optimal solution with deterministic ownership.*

Proof. Let φ be any feasible allocation rule for (A.13), and let $G \in \mathcal{N}$ denote the (possibly random) owner induced by φ (which is publicly observed before actions are chosen). Let $\psi(a,g) = p(\theta)\varphi(a,g \mid \theta)$ denote the join distribution of (a,g,θ) induced by φ. For each g with

$\mathbb{P}_\psi(G = g) > 0$, define the conditional decision rule $\varphi^g(\cdot \mid \theta) \in \Delta(\mathcal{A})$ by

$$\varphi^g(a \mid \theta) \equiv \mathbb{P}_\psi(a \mid \theta,\, G = g) = \frac{\psi(a, g, \theta)}{\sum_a \psi(a, g, \theta)}.$$

Because parties observe $G = g$ before choosing actions, obedience of φ implies that, conditional on $G = g$, the induced rule φ^g satisfies the obedience constraints under ownership g. Hence the conditional expected surplus cannot exceed $TS(g)$:

$$\mathbb{E}_\psi[TS(a, G; \theta) \mid G = g] \leqslant TS(g).$$

Taking expectations over G yields

$$\mathbb{E}_\psi[TS(a, G; \theta)] = \sum_{g \in \mathcal{N}} \Pr(G = g)\, \mathbb{E}_\psi[TS \mid G = g] \;\leqslant\; \sum_{g \in \mathcal{N}} \Pr(G = g)\, TS(g) \;\leqslant\; \max_{g \in \mathcal{N}} TS(g).$$

Finally, choosing $G \equiv g^* = \arg\max_g TS(g)$ and using an optimal obedient recommendation rule that attains $TS(g^*)$ achieves the upper bound. This proves that an optimal solution exists with deterministic ownership. □

The solution to the relaxed (obedience-only) problem in (2.2) with deterministic ownership provides an upper bound for the mechanism-design problem (A.11). Let σ^* be an optimal decision rule for the problem in (2.2), and let g^* denote an optimal owner, $g^* \in \arg\max_g TS(g)$.

Consider the following contract (φ, t):

$$\varphi(a, g \mid \theta) = \begin{cases} \sigma^*(a \mid \theta) & \text{if } g = g^* \\ 0 & \text{if } g \neq g^* \end{cases} \quad \text{for all } (a, \theta) \in \mathcal{A} \times \Theta \tag{A.14}$$

$$t_i(a, g, \theta) = t_i = \underline{U}_i - \mathbb{E}_\psi\left[u_i(a, g; \theta)\right] \quad \text{for } i \neq 1 \text{ and all } (a, g, \theta) \in \mathcal{A} \times \mathcal{N} \times \Theta \tag{A.15}$$

$$t_1(a, g, \theta) = -\sum_{i=2}^{N} t_i \tag{A.16}$$

This contract assigns ownership deterministically to g^* and induces the same distribution over (a, θ) as the optimal obedient rule σ^* that solves (2.2). In Section A.1, we characterize the optimal decision rule (Proposition 1) and show that it attains the maximum expected surplus in (2.2), thereby establishing an upper bound for (A.11). Next, we show that the contract (A.14)–(A.16) satisfies the incentive-compatibility constraints and therefore achieves this upper bound.

Lemma 4 (Incentive Compatibility)**.** *Fix the optimal owner g^* and the nested obedient rule σ^**

under g^*. The direct mediated contract (A.14)–(A.16) *based on* (g^*, σ^*) *is incentive compatible.*

Proof. Transfers are constant under (A.15)–(A.16) and hence cancel from all report and deviation comparisons. Fix a party i, a true type θ_i, an alternative report $\hat{\theta}_i \in \Theta_i$, and an arbitrary action-stage deviation $\delta_i : D_i \to \mathcal{A}_i$.

Step 1: Upper Bound for a Single Task. Fix any target $j \neq i$. Under truthful reporting by party j, we have $\hat{\theta}_j = \theta_j$, and party i receives a recommendation

$$x_{ij} = \bar{x}_{ij} = \begin{cases} \theta_j, & \xi \leqslant \tau_{ij}^{\text{safe}}(g^*), \\ -\theta_j, & \xi > \tau_{ij}^{\text{safe}}(g^*). \end{cases} \tag{A.17}$$

In Lemma 1, we proved that the posterior belief that $\theta_j = x_{ij}$ under (A.17) is equal to $\tau_{ij}^{\text{safe}}(g^*)$. Since $\tau_{ij}^{\text{safe}}(g^*) \geqslant \frac{1}{2}$, x_{ij} is a posterior mode for θ_j given d_i. Thus, among all choices of i's productive action for target j, the maximal achievable success probability in task (i, j) equals $\tau_{ij}^{\text{safe}}(g^*)$. Therefore, for any deviation δ_i and any report $\hat{\theta}_i$, the probability that task (i, j) in project k succeeds (event $C_{k,(ij)}$ defined in (A.6)) has the following upper bound:

$$\mathbb{P}(C_{k,(ij)} \mid \hat{\theta}_i, \delta_i, \theta_i) \;\leqslant\; \tau_{ij}^{\text{safe}}(g^*) \qquad \text{for all } j \neq i. \tag{A.18}$$

A similar bound applies to tasks performed by *other* parties who obey. Fix $\ell \neq i$ and $j \neq \ell$. Since ℓ is held to obey equilibrium recommendations, its match success satisfies

$$\mathbb{P}(C_{k,(\ell j)} \mid \hat{\theta}_i, \delta_i, \theta_i) = \begin{cases} \tau_{\ell j}^{\text{safe}}(g^*), & j \neq i \text{ (since } \hat{\theta}_j = \theta_j), \\ \tau_{\ell i}^{\text{safe}}(g^*), & j = i,\ \hat{\theta}_i = \theta_i, \\ 1 - \tau_{\ell i}^{\text{safe}}(g^*), & j = i,\ \hat{\theta}_i = -\theta_i. \end{cases}$$

Furthermore, since $\tau_{\ell i}^{\text{safe}}(g^*) \geqslant \frac{1}{2}$, in all cases we have

$$\mathbb{P}(C_{k,(\ell j)} \mid \hat{\theta}_i, \delta_i, \theta_i) \leqslant \tau_{\ell j}^{\text{safe}}(g^*). \tag{A.19}$$

Step 2: Upper Bound on Project Success. For each project k, success requires all required tasks to succeed: $E_k = \bigcap_{(\ell,j) \in \mathcal{Q}_k(g^*)} C_{k,(\ell j)}$. Hence for any $\hat{\theta}_i, \delta_i$,

$$\mathbb{P}(E_k \mid \hat{\theta}_i, \delta_i, \theta_i) \leqslant \min_{(\ell,j) \in \mathcal{Q}_k(g^*)} \mathbb{P}(C_{k,(\ell j)} \mid \hat{\theta}_i, \delta_i, \theta_i) \leqslant \min_{(\ell,j) \in \mathcal{Q}_k(g^*)} \tau_{\ell j}^{\text{safe}}(g^*).$$

Under truthful reporting and obedience, each required task (ℓ, j) succeeds if and only if $\xi \leqslant$

$\tau_{\ell j}^{\text{safe}}(g^*)$, so (because the same ξ is shared across tasks)

$$\mathbb{P}(E_k \mid \text{truth, obey}, \theta_i) = \min_{(\ell,j)\in\mathcal{Q}_k(g^*)} \tau_{\ell j}^{\text{safe}}(g^*).$$

That is, truthful reporting and obedience attain the upper bound for every project.

Since $u_i^c = \sum_{k\in\mathcal{K}} \alpha_{ik} V_k \mathbf{1}\{E_k\}$ with $\alpha_{ik} V_k \geqslant 0$, maximizing each $\mathbb{P}(E_k)$ implies that truthful reporting and obedience maximize i's expected cooperative payoff.

No Expropriation. Under σ^* the posterior accuracy about any target j is capped at $\tau_{ij}^{\text{safe}}(g^*)$, so by the definition of the safety threshold (see Lemma 1), any expropriation attempt yields a weakly lower expected payoff than $\zeta_{ij} = 0$. The expropriation payoff is unaffected by i's report because i's information about θ_{-i} is independent of $\hat{\theta}_i$.

Conclusion. Combining the cooperative and expropriation parts yields the double-deviation IC constraints (A.9). □

The derivations in this section prove the following proposition:

Proposition 3. *Let $g^* \in \arg\max_{g\in\mathcal{N}} TS(g)$ and let σ^* be an optimal obedient decision rule for the omniscient-mediator problem under owner g^* (so σ^* attains $TS(g^*)$ in* (2.2)*). Consider the direct mediated contract (φ, t) defined by* (A.14)–(A.16)*. This contract solves the mechanism-design program* (A.11)*.*

Proof. The relaxed problem (A.13) provides an upper bound to (A.11). Lemma 3 shows that the relaxed problem has a solution with deterministic ownership. Thus, there is a solution to the relaxed problem that coincides with a solution the omniscient mediator's problem in (2.2). In the contract based on the solution to the mediator's problem, the transfers are constructed in a way to satisfy the participation constraints, and Lemma 4 shows that this contract is incentive compatible, which completes the proof. □

A.3 Contractible Actions and Cooperative Payoffs

In the baseline model, all components of the action plan $a_i = (x_i, \bar{x}_i, z_i, \bar{z}_i) \in \mathcal{A}_i$ are *non-contractible*: the mediator can only send private recommendations, and obedience must be verified against deviations in *all* components. In addition, project shares $\{\alpha_{ik}\}_{i,k}$ are fixed and exogenously given.

This section studies two counterfactual environments. First, we allow expropriation actions to be contractible; this makes it possible to implement the first best. Second, we allow cooperative actions and project shares to be contractible. We show that making cooperative

actions and shares contractible does not increase the maximum attainable surplus relative to the baseline.

For notational convenience, decompose each party's action plan into a *cooperation block* and an *expropriation block*:

$$a_i^c \equiv (x_i, \bar{x}_i) \in \mathcal{A}_i^c = \{-1, 1\}^{2N} \qquad a_i^e \equiv (z_i, \bar{z}_i) \in \mathcal{A}_i^e = \{-1, 0, 1\}^{2(N-1)},$$

so that $\mathcal{A}_i = \mathcal{A}_i^c \times \mathcal{A}_i^e$ and $a_i = (a_i^c, a_i^e)$. The mediator observes θ and draws an action profile $a = (a_i)_{i \in \mathcal{N}} \in \mathcal{A}$ from a direct decision rule $\sigma(\cdot \mid \theta) \in \Delta(\mathcal{A})$.

(i) Contractible expropriation actions. Suppose the mediator can *enforce* expropriation actions $(z, \bar{z})$, while cooperative actions $(x, \bar{x})$ remain non-contractible. Then, after receiving a recommendation $a_i = (a_i^c, a_i^e)$, party i can deviate only in the cooperation block a_i^c; the expropriation block a_i^e is fixed.

Accordingly, σ is *obedient under g with contractible expropriation* if, for every $i \in \mathcal{N}$, every $\theta_i \in \Theta_i$, every recommended $a_i = (a_i^c, a_i^e) \in \mathcal{A}_i$, and every alternative $\tilde{a}_i^c \in \mathcal{A}_i^c$,

$$\sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma(a_i, a_{-i} \mid \theta_i, \theta_{-i}) \Big[u_i\big(((a_i^c, a_i^e), a_{-i}), g; \theta\big) - u_i\big(((\tilde{a}_i^c, a_i^e), a_{-i}), g; \theta\big) \Big] \geqslant 0. \tag{A.20}$$

Define the corresponding omniscient benchmark value:

$$TS^e(g) \equiv \max_{\sigma} \sum_{\theta} p(\theta) \sum_{a} \sigma(a \mid \theta) \cdot TS(a, g; \theta) \tag{A.21}$$
$$\text{subject to (A.20)}, \quad \sigma(a \mid \theta) \geqslant 0, \quad \sum_{a} \sigma(a \mid \theta) = 1 \text{ for all } \theta \in \Theta$$

(ii) Contractible actions and cooperative payoffs. Suppose instead that the mediator can *enforce* cooperative actions $(x, \bar{x})$, while expropriation actions $(z, \bar{z})$ remain non-contractible. Then party i can deviate only in the expropriation block a_i^e; the cooperation block a_i^c is fixed.

Accordingly, σ is *obedient under g with contractible cooperation* if, for every $i \in \mathcal{N}$, every $\theta_i \in \Theta_i$, every recommended $a_i = (a_i^c, a_i^e) \in \mathcal{A}_i$, and every alternative $\tilde{a}_i^e \in \mathcal{A}_i^e$,

$$\sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{a_{-i} \in \mathcal{A}_{-i}} \sigma(a_i, a_{-i} \mid \theta_i, \theta_{-i}) \Big[u_i\big(((a_i^c, a_i^e), a_{-i}), g; \theta\big) - u_i\big(((a_i^c, \tilde{a}_i^e), a_{-i}), g; \theta\big) \Big] \geqslant 0. \tag{A.22}$$

Suppose the mediator can also choose and enforce the division of cooperative payoffs $\{\alpha_{ik}\}_{i,k}$.

Thus, the corresponding omniscient mediator's benchmark value is

$$TS^c(g) \equiv \max_{\sigma} \sum_{\theta} p(\theta) \sum_{a} \sigma(a \mid \theta) \cdot TS(a, g; \theta) \tag{A.23}$$

$$\text{s.t. (A.22)}, \quad \alpha_{ik} \geqslant 0, \quad \sum_{i \in \mathcal{N}} \alpha_{ik} = 1 \text{ for all } k \in \mathcal{K}$$

$$\sigma(a \mid \theta) \geqslant 0, \quad \sum_{a} \sigma(a \mid \theta) = 1 \text{ for all } \theta \in \Theta$$

Proposition 4 (Contractability and the attainable surplus)**.** *Fix any ownership assignment $g \in \mathcal{N}$. If expropriation actions $(z, \bar{z})$ are contractible, then*

$$TS^e(g) = V_{\text{total}} \qquad \textit{for every } g \in \mathcal{N}.$$

If cooperative actions $(x, \bar{x})$ and projects' shares $\{\alpha_{ik}\}_{i,k}$ are contractible, then

$$TS^c(g) = TS(g) \qquad \textit{for every } g \in \mathcal{N},$$

where $TS(g)$ is the value characterized in Proposition 1.

Proof. **(i) Contractible expropriation actions:** $TS^e(g) = V_{\text{total}}$. Define a (deterministic) decision rule σ^{FB} as follows. For every $\theta \in \Theta$:

1. Set *all* expropriation actions to zero: $z_{ij} = 0$ and $\bar{z}_{ij} = 0$ for all i and all $j \neq i$;
2. Recommend $x_{ij} = \bar{x}_{ij} = \theta_j$ for all $i \in \mathcal{N}$ and $j \in \mathcal{N}$.

Under σ^{FB}, there is no expropriation, and every project succeeds surely. Thus the total surplus is V_{total}. Since $(z, \bar{z})$ are enforced, party i can deviate only in $a_i^c = (x_i, \bar{x}_i)$. If i deviates to some $\tilde{a}_i^c \neq a_i^c$, then at least one matching requirement that uses i's action is violated, which can only weakly decrease i's cooperative payoff (since $\alpha_{ik} \geqslant 0$ for all k). Thus, σ^{FB} is obedient.

(ii) Contractible cooperation and shares: $TS^c(g) = TS(g)$. First, we show that $TS^c(g) \geqslant TS(g)$. Let σ^g be the nested no-expropriation decision rule constructed in the proof of Proposition 1 (see (A.2)-(A.3)). Under σ^g we have $\zeta_{ij} = 0$ for all $i \neq j$ almost surely, and each project k succeeds with probability $\min_{(i,j) \in \mathcal{Q}_k(g)} \tau_{ij}^{\text{safe}}(g)$, so it achieves $TS(g)$. This decision rule is obedient and satisfies (A.22).

Second, we show that $TS^c(g) \leqslant TS(g)$. This inequality follows from the same steps as in the proof of Lemma 2. The binding restriction comes from the expropriation obedience constraints. Since expropriation actions remain non-contractible, the same upper bound continues to apply even when cooperative actions are contractible.

Finally, allowing the mediator to choose the shares $\{\alpha_{ik}\}$ does not affect the objective W, which depends only on total project values. It also does not affect the obedience constraints (A.22): because cooperative actions are fixed in both terms, cooperative payoffs cancel when comparing deviations in expropriation. Hence, making shares contractible is immaterial for the welfare bound. □

A.4 Optimal Decision Rule Under Arbitrary Expropriation Costs

We describe the results as follows. First, for any obedient decision rule we derive an upper bound on the total surplus. Second, we construct an obedient decision rule that attains this bound. Finally, we formulate the mediator's optimization problem.

A.4.1 The Upper Bound on Total Surplus

Information Accuracy. Fix an owner $g \in \mathcal{N}$. Consider an obedient decision rule $\sigma \in \Sigma$ with the induced joint distribution $\psi(a, \theta) = p(\theta)\sigma(a \mid \theta)$ over states and actions. Let $I_i = (\theta_i, a_i)$ be party i's private information at the action stage, and let

$$q_{ij}(I_i) \equiv \max_{\omega \in \{-1,1\}} \mathbb{P}_\psi(\theta_j = \omega \mid I_i)$$

denote the maximum conditional accuracy of matching state θ_j.

The conditional accuracy $q_{ij}(I_i)$ provides an upper bound on the probability that party i successfully matches task j. Let $\mathcal{Q}_{k,i}(g) \equiv \{j : (i,j) \in \mathcal{Q}_k(g)\}$ denote the set of targets party i must match for project k to succeed. Let $E_{k,i}$ denote the event that party i succeeds in all its tasks in project k. Since this event requires *all* tasks to succeed, we have

$$\mathbb{P}_\psi\left(E_{k,i}\right) = \mathbb{E}_\psi\left[\mathbb{P}_\psi(E_{k,i} \mid I_i)\right] \leqslant \mathbb{E}_\psi\left[\min_{j \in \mathcal{Q}_{k,i}(g)} \{q_{ij}(I_i)\}\right].$$

Expropriation Attempt Patterns. For each armed party $i \in \mathrm{Arm}(g)$, we define the random expropriation attempt pattern S_i and its probability π_{iS} as

$$S_i \equiv \{j \in \mathcal{N} \backslash \{i\} : \zeta_{ij} \neq 0\}, \quad \pi_{iS} \equiv \mathbb{P}_\psi\left(S_i = S\right)$$

That is, S_i is the set of targets j for which i is recommended to attempt expropriation. Let $J_i(g) \equiv \bigcup_k \mathcal{Q}_{k,i}(g)$ denote the set of all production targets for party i. Without loss of optimality, we can focus on decision rules in which i receives no expropriation recommendations for irrelevant targets: $\zeta_{ij} = 0$ for all $j \notin J_i(g)$, which implies $S_i \subseteq J_i(g)$.

Upper Bound on a Party's Success. We derive an upper bound on party i's success probability by conditioning on the attempt pattern S_i. By the Law of Iterated Expectations and Jensen's inequality (as the minimum function is concave), we have:

$$\mathbb{P}_\psi\left(E_{k,i}\right) \leqslant \mathbb{E}_S\left[\mathbb{E}_\psi\left[\min_{j\in\mathcal{Q}_{k,i}(g)} q_{ij}(I_i) \;\middle|\; S_i\right]\right] \leqslant \mathbb{E}_S\left[\min_{j\in\mathcal{Q}_{k,i}(g)} \mathbb{E}_\psi\left[q_{ij}(I_i) \mid S_i\right]\right]. \tag{A.24}$$

The obedience constraints impose the following bounds on the accuracy $q_{ij}(I_i)$. If party i's recommendation is safe action $\zeta_{ij} = 0$ (i.e., $j \notin S_i$), obedience requires $q_{ij}(I_i) \leqslant \tau_{ij}^{\text{safe}}(g)$. Conversely, if i is recommended expropriation $\zeta_{ij} \neq 0$ (i.e., $j \in S_i$), it must be sufficiently certain about the target state: $q_{ij}(I_i) \geqslant \tau_{ij}^{\text{safe}}(g)$. Thus, $\mathbb{E}_\psi[q_{ij}(I_i) \mid S_i] \leqslant \tau_{ij}^{\text{safe}}(g)$ for $j \notin S_i$ and $\mathbb{E}_\psi[q_{ij}(I_i) \mid S_i] \geqslant \tau_{ij}^{\text{safe}}(g)$ for $j \in S_i$.

To derive the upper bound, we set the accuracy of safe actions to their maximum permissible value. We define the conditional accuracy variable m_{ij}^S as follows:

$$m_{ij}^S \equiv \begin{cases} \tau_{ij}^{\text{safe}}(g) & \text{if } j \notin S \quad \text{(Safe Action)}, \\ \mathbb{E}_\psi[q_{ij}(I_i) \mid S_i] \in [\tau_{ij}^{\text{safe}}(g), 1] & \text{if } j \in S \quad \text{(Expropriation Attempt)}. \end{cases}$$

Combining the derived inequalities with these definitions, we obtain the following upper bound on the probability that party i succeeds in project k:

$$\mathbb{P}_\psi\left(E_{k,i}\right) \leqslant \sum_{S\subseteq J_i(g)} \pi_{iS}\, \rho_{k,i}^S \equiv P_{k,i}(\pi, m; g), \quad \rho_{k,i}^S \equiv \min_{j\in\mathcal{Q}_{k,i}(g)} m_{ij}^S,$$

where we adopt the convention $\min \varnothing \equiv 1$.

The Party's Success Across Projects. If party i participates in multiple projects, the success events for different projects can be correlated. For any nonempty subset $L \subseteq \mathcal{K}$, define the event that party i succeeds in *all* projects in L: $E_{L,i} \equiv \bigcap_{k\in L} E_{k,i}$. If i succeeds in all projects in L, it must match *every* target in the union $\bigcup_{k\in L} \mathcal{Q}_{k,i}(g)$. Hence, repeating the previous argument in (A.24) with the target set $\bigcup_{k\in L} \mathcal{Q}_{k,i}(g)$ gives

$$\mathbb{P}_\psi(E_{L,i}) \leqslant \sum_{S\subseteq J_i(g)} \pi_{iS} \min_{j\in\cup_{k\in L}\mathcal{Q}_{k,i}(g)} m_{ij}^S = \sum_{S\subseteq J_i(g)} \pi_{iS} \min_{k\in L} \rho_{k,i}^S. \tag{A.25}$$

Up to this point, we have showed that any obedient decision rule σ (with induced distribution ψ) is associated with two statistics: expropriation attempt probabilities π and conditional expected accuracies m. These statistics pin down the upper bound on the probability of i's success in various projects. Specifically, equation (A.25) provides a restriction on the joint distribution of the random vector $(\mathbf{1}\{E_{k,i}\})_{k\in\mathcal{K}}$ under ψ.

To upper bound cooperative surplus, we replace the unknown conditional joint law of $(E_{k,i})_k$ by an *auxiliary* joint law that attains all bounds in (A.25) simultaneously. Let $\xi_i \sim \mathrm{Unif}[0,1]$ be an auxiliary independent random variable. Conditional on $S_i = S$, define auxiliary success events $\widetilde{E}^S_{k,i} \equiv \{\xi_i \leqslant \rho^S_{k,i}\}$ for $k \in \mathcal{K}$. Then, for every nonempty $L \subseteq \mathcal{K}$,

$$\mathbb{P}\Big(\bigcap_{k\in L} \widetilde{E}^S_{k,i}\Big) = \Pr\Big(\xi_i \leqslant \min_{k\in L} \rho^S_{k,i}\Big) = \min_{k\in L} \rho^S_{k,i},$$

so the auxiliary construction saturates the right-hand side of (A.25) for all L at once.

Now mix over attempt patterns using the *actual* weights $\pi_{iS} \equiv \mathbb{P}_\psi(S_i = S)$: first draw S_i according to π_i, then draw $\xi_i \sim \mathrm{Unif}[0,1]$, and set $\widetilde{E}_{k,i} \equiv \widetilde{E}^{S_i}_{k,i}$. By the law of total probability, for every $L \subseteq \mathcal{K}$, we have

$$\mathbb{P}\Big(\bigcap_{k\in L} \widetilde{E}_{k,i}\Big) = \sum_{S\subseteq J_i(g)} \pi_{iS} \Pr\Big(\bigcap_{k\in L} \widetilde{E}^S_{k,i}\Big) = \sum_{S\subseteq J_i(g)} \pi_{iS} \min_{k\in L} \rho^S_{k,i}.$$

Since cooperative surplus is increasing in each party's project-success indicators, replacing the unknown joint law of $(E_{k,i})_k$ by the "maximal" law $(\widetilde{E}_{k,i})_k$ (which saturates all upper bounds) can only increase the best cooperative surplus consistent with (A.25). Thus, evaluating the cooperative part under $(\widetilde{E}_{k,i})$ yields an upper bound. Later we show that this upper bound is attainable.

Define the auxiliary (maximal) success type of party i as the random set

$$T_i \equiv \{k \in \mathcal{K} : \ \widetilde{E}_{k,i} \text{ occurs}\} \subseteq \mathcal{K}.$$

Let $\beta_{iT}(\pi, m; g)$ denote the marginal distribution of T_i:

$$\beta_{iT}(\pi, m; g) \equiv \mathbb{P}(T_i = T), \qquad T \subseteq \mathcal{K}.$$

Conditional on $S_i = S$, the event $\{T_i = T\}$ is equivalent to

$$\{\xi_i \leqslant \rho^S_{k,i} \ \forall k \in T\} \cap \{\xi_i > \rho^S_{k,i} \ \forall k \notin T\} \quad \Longleftrightarrow \quad \xi_i \in \Big(\max_{k\notin T} \rho^S_{k,i}, \ \min_{k\in T} \rho^S_{k,i}\Big],$$

with $\max \varnothing \equiv 0$ and $\min \varnothing \equiv 1$. Since $\xi_i \sim \mathrm{Unif}[0,1]$,

$$\mathbb{P}(T_i = T \mid S_i = S) = \Big[\min_{k\in T} \rho^S_{k,i} - \max_{k\notin T} \rho^S_{k,i}\Big]_+,$$

and therefore

$$\beta_{iT}(\pi, m; g) = \sum_{S\subseteq J_i(g)} \pi_{iS} \Big[\min_{k\in T} \rho^S_{k,i} - \max_{k\notin T} \rho^S_{k,i}\Big]_+. \tag{A.26}$$

Fix (π, m) and hence the induced marginals $\big(\beta_{iT}(\pi, m; g)\big)_{T \subseteq \mathcal{K}}$ for each active party i. At this stage, the mediator still has freedom to choose how to *correlate* the success types $(T_i)_i$ across parties. Let $\widetilde{\mathcal{P}}_k(g)$ denote the set of participants in project k (those whose matching tasks appear in $\mathcal{Q}_k(g)$).[4] Project k succeeds if and only if every $i \in \widetilde{\mathcal{P}}_k(g)$ succeeds in k, which in type language is

$$\text{Project } k \text{ succeeds} \quad \Longleftrightarrow \quad k \in T_i \;\; \text{for all } i \in \widetilde{\mathcal{P}}_k(g).$$

Let $\mathcal{I}(g) \equiv \bigcup_{k \in \mathcal{K}} \widetilde{\mathcal{P}}_k(g)$ be the set of parties whose actions matter for at least one project. A *coupling* is a joint distribution

$$\gamma \in \Delta\Big((2^{\mathcal{K}})^{\mathcal{I}(g)}\Big) \quad \text{over type profiles} \quad \mathbf{T} = (T_i)_{i \in \mathcal{I}(g)}.$$

The only feasibility restrictions on γ are the marginal constraints

$$\sum_{\mathbf{T}_{-i}} \gamma(\mathbf{T}) = \beta_{iT_i}(\pi, m; g) \qquad \forall i \in \mathcal{I}(g),\ \forall T_i \subseteq \mathcal{K},$$

because (i) each party observes only its own recommendation, so obedience is pinned down by its own marginal signal structure (already encoded in (π, m)), and (ii) the mediator can implement any cross-party correlation by correlating the latent randomization used to generate success events.

Under a coupling γ, cooperative surplus equals

$$\sum_{k \in \mathcal{K}} V_k \, \mathbb{P}_\gamma\big(k \in T_i \;\forall i \in \widetilde{\mathcal{P}}_k(g)\big) = \sum_{k \in \mathcal{K}} V_k \sum_{\mathbf{T}} \gamma(\mathbf{T}) \, \mathbf{1}\{k \in T_i \;\forall i \in \widetilde{\mathcal{P}}_k(g)\}.$$

Therefore, for fixed (π, m), the mediator's *best* cooperative surplus is the value of the following *multi-marginal optimal transport (MMOT)* linear program:

$$\begin{aligned} \mathrm{W}\big(\pi, m; g\big) \equiv \max_{\gamma \geqslant 0} \; & \sum_{k \in \mathcal{K}} V_k \sum_{\mathbf{T}} \gamma(\mathbf{T}) \, \mathbf{1}\{k \in T_i \;\forall i \in \widetilde{\mathcal{P}}_k(g)\} \\ \text{s.t.} \quad & \sum_{\mathbf{T}_{-i}} \gamma(\mathbf{T}) = \beta_{iT_i}(\pi, m; g) \;\; \forall i, T_i. \end{aligned} \tag{A.27}$$

The same parameters (π, m) also pin down probabilities of successful and unsuccessful

[4] The set $\widetilde{\mathcal{P}}_k(g)$ differs from the set of *active* participants $\mathcal{P}_k(g)$ defined in (3.2) in the main text by including parties with trivial tasks: $i \in \widetilde{\mathcal{P}}_k(g)$ for $(i, i) \in \mathcal{Q}_k(g)$.

expropriation attempts for $i \in \mathrm{Arm}(g)$:

$$\mathbb{P}_\psi(\zeta_{ij} = \theta_j) = \sum_{S: j \in S} \pi_{iS}\, m_{ij}^S, \qquad \mathbb{P}_\psi(\zeta_{ij} = -\theta_j) = \sum_{S: j \in S} \pi_{iS}\, (1 - m_{ij}^S)$$

Hence, combining the cooperative bound (A.27) with the surplus contribution of expropriation yields the following *general* upper bound on expected total surplus under owner g:

$$\begin{aligned} \mathbb{E}_\psi[TS(a; g, \theta)] \;\leqslant\; & \mathrm{W}\big(\pi, m; g\big) - \\ & - \sum_{i \in \mathrm{Arm}(g)} \sum_{j \in J_i(g) \setminus \{i\}} \Big[(\lambda_{ij} - b_{ij}) \sum_{S: j \in S} \pi_{iS}\, m_{ij}^S + c_{ij} \sum_{S: j \in S} \pi_{iS}\, (1 - m_{ij}^S) \Big]. \end{aligned} \tag{A.28}$$

A.4.2 Attainability

Fix any feasible (π, m) and an optimal coupling γ that solves (A.27). We construct an obedient decision rule that achieves the upper bound in (A.28).

Step 1: Project-Success Regions. For each $i \in \mathrm{Arm}(g)$, choose a measurable partition of $[0, 1]$ into intervals $(I_{iS})_{S \subseteq J_i(g)}$ with $|I_{iS}| = \pi_{iS}$. For each project $k \in K$, define the region

$$\mathcal{S}_{k,i} \equiv \bigcup_{S \subseteq J_i(g)} \Big(I_{iS} \times [0, \rho_{k,i}^S] \Big) \subseteq [0, 1]^2.$$

For each subset $T \subseteq K$ define the corresponding cell in the induced partition of $[0, 1]^2$:

$$\Delta_{i,T} \equiv \left(\bigcap_{k \in T} \mathcal{S}_{k,i} \right) \cap \left(\bigcap_{k \notin T} \mathcal{S}_{k,i}^c \right).$$

Then $\{\Delta_{i,T}\}_{T \subseteq K}$ is a measurable partition of $[0, 1]^2$, and the Lebesgue measure of each region is $\mathrm{Leb}(\Delta_{i,T}) = \beta_{iT}(\pi, m; g)$ by the definition of β_{iT}.

Step 2: Implement the chosen coupling γ. Draw $(T_i)_{i \in \mathcal{I}(g)}$ according to γ. Conditional on $(T_i)_i$, draw (η_i, ξ_i) independently across i with

$$(\eta_i, \xi_i) \;\sim\; \mathrm{Unif}\big(\Delta_{i,T_i}\big).$$

Because each T_i is selected with probability equal to the area of Δ_{i,T_i}, this implies:

1. *Uniform marginals:* unconditionally, (η_i, ξ_i) is uniform on $[0, 1]^2$ for each i.
2. *Type realization:* for each $k \in K$, we have $k \in T_i \;\Leftrightarrow\; (\eta_i, \xi_i) \in \mathcal{S}_{k,i}$.

Step 3: Send Recommendations. Fix a realized state θ. For each disarmed party $i \notin \mathrm{Arm}(g)$, recommend no expropriation and fully revealing matching actions for all relevant targets (so such parties match surely and never expropriate). Now fix $i \in \mathrm{Arm}(g)$. Define the realized attempt pattern from η_i:

$$S_i = S \quad \Longleftrightarrow \quad \eta_i \in I_{iS}.$$

For each $j \in J_i(g)$ define the recommended sign

$$s_{ij} \equiv \begin{cases} \theta_j, & \xi_i \leqslant m_{ij}^{S_i}, \\ -\theta_j, & \xi_i > m_{ij}^{S_i}. \end{cases}$$

Recommend cooperative actions $x_{ij} = \bar{x}_{ij} = s_{ij}$ for all $j \in J_i(g)$, and recommend expropriation

$$\zeta_{ij} = \begin{cases} s_{ij}, & j \in S_i, \\ 0, & j \notin S_i. \end{cases}$$

For targets not in $J_i(g)$, recommend uninformative (constant) cooperative actions $x_{ij} = \bar{x}_{ij} = 1$ and $\zeta_{ij} = 0$.

Posteriors. The information available to party $i \in \mathrm{Arm}(g)$ about targets $j \neq i$ before taking actions is the attempted set S_i (encoded in $\zeta_{ij} \neq 0$) and the recommended signs $s_i \equiv (s_{ij})_{j \in J_i(g)}$ (encoded in $(x_{ij}, \bar{x}_{ij})$). The attempted set S_i is independent of θ_i. Indeed, $(T_i)_i \sim \gamma$ is drawn independently of θ, and (η_i, ξ_i) are drawn from Δ_{i,T_i} independently of θ.

Fix a realized message (S_i, s_i) and $j \in J_i(g)$. Compute the following posterior:

$$\mathbb{P}\left(\theta_j = s_{ij} \mid S_i, s_i\right) = \mathbb{E}\left[\mathbb{P}\left(\theta_j = s_j \mid S_i, s_i, \xi_i\right) \mid S_i, s_i\right]$$

Conditional on (S_i, ξ_i), we have $\theta_j = s_{ij}$ if an only if $\xi_i \leqslant m_{ij}^{S_i}$, and the map $\theta_{J_i(g)} \mapsto s_i$ is a deterministic coordinate-wise sign flip bijection on $\{-1, 1\}^{J_i(g)}$. Since $\theta_{J_i(g)}$ is uniform on $\{-1, 1\}^{J_i(g)}$, it follows that

$$\mathbb{P}(s_i = \bar{s} \mid S_i, \xi_i) = 2^{-|J_i(g)|} \quad \text{for every } \bar{s} \in \{-1, 1\}^{J_i(g)},$$

which does not depend on ξ_i. Hence, s_i is independent of ξ_i conditional on S_i. Using this independence, we get

$$\mathbb{P}(\theta_j = s_{ij} \mid S_i, s_i) = \mathbb{P}(\xi_i \leqslant m_{ij}^{S_i} \mid S_i, s_i) = \mathbb{P}(\xi_i \leqslant m_{ij}^{S} \mid S_i) = m_{ij}^{S_i},$$

where the last equality uses $\xi_i \mid S_i \sim \text{Unif}[0,1]$. Therefore, the posterior for target j is

$$\mathbb{P}(\theta_j = s_{ij} \mid \theta_i, S_i, s_i) = m_{ij}^{S_i}, \qquad P(\theta_j = -s_{ij} \mid \theta_i, S_i, s_i) = 1 - m_{ij}^{S_i}. \tag{A.29}$$

Obedience. Fix $i \in \text{Arm}(g)$. If $j \in S_i$, the recommended expropriation is ζ_{ij}. Since $m_{ij}^{S_i} \geqslant \tau_{ij}^{\text{safe}} = \tau_{ij}^{\star}$, the posterior in (A.29) implies that it is optimal for party i to follow the recommendation. If $j \notin S_i$, then $m_{ij}^{S_i} = \tau_{ij}^{\text{safe}}$, and it is optimal for i to follow the recommendation $\zeta_{ij} = 0$.

The obedience in cooperative actions follows the same logic as in the proof of Lemma 1. Since $\xi_i \mid S_i \sim \text{Unif}[0,1]$ and $m_{ij}^{S_i} \geqslant \tau_{ij}^{\text{safe}}(g) > \frac{1}{2}$, following the cooperative recommendation ensures that a project succeeds with probability higher than $\frac{1}{2}$, while deviating decreases this probability to less than $\frac{1}{2}$ in each project where i's actions are required, and i deviates.

A.4.3 The Mediator's Problem

Up to this point, we have shown that each obedient decision rule has an upper bound on the total surplus in (A.28), and this bound is attainable. Now, we formulate the mediator's problem using this upper bound.

Let $\mathcal{PM}(g)$ denote the feasible set for the expropriation attempt probabilities π and corresponding accuracies m. That is, $\pi \equiv (\pi_1, \ldots, \pi_N)$, where $(\pi_{iS})_{S \subseteq \mathcal{N}}$, $\pi_{iS} \geqslant 0$, and $\sum_{S \subseteq \mathcal{N}} \pi_{iS} = 1$. For accuracies, $m \equiv (m_1, \ldots, m_N)$, where $m_i \equiv (m_{ij}^S)_{j \in \mathcal{N}, S \subseteq \mathcal{N}}$, $m_{ij}^S = \tau_{ij}^{\text{safe}}(g)$ for $j \notin S$, and $m_{ij}^S \in [\tau_{ij}^{\text{safe}}(g), 1]$ for $j \in S$.

The derivations in this section prove the following proposition:

Proposition 5. *Suppose that successful expropriation has arbitrary costs $\lambda_{ij} - b_{ij} > 0$. The maximum attainable total surplus under owner g is*

$$\begin{aligned} TS(g) = \max_{(\pi,m) \in \mathcal{PM}(g)} \Big\{ & \mathrm{W}(\pi, m; g) \\ & - \sum_{i \in \text{Arm}(g)} \sum_{j \in J_i(g) \setminus \{i\}} \Big[(\lambda_{ij} - b_{ij}) \sum_{S: j \in S} \pi_{iS}\, m_{ij}^S + c_{ij} \sum_{S: j \in S} \pi_{iS}\, (1 - m_{ij}^S) \Big] \Big\}, \end{aligned} \tag{A.30}$$

where $\mathrm{W}(\pi, m; g)$ is a solution to the multi-marginal optimal transport problem in (A.27)*.*

The problem in (A.30) can be simplified under the following assumption.

Assumption 4. *For each $g \in \mathcal{N}$, the family of project task sets $\{\mathcal{Q}_k(g)\}_{k \in \mathcal{K}}$ is totally ordered by inclusion. That is, the projects can be indexed $k = 1, \ldots, |\mathcal{K}|$ such that:*

$$\mathcal{Q}_1(g) \supseteq \mathcal{Q}_2(g) \supseteq \cdots \supseteq \mathcal{Q}_{|\mathcal{K}|}(g).$$

This assumption implies that projects can be ranked by difficulty: Project 1 is the "hardest" (requires the most tasks), while Project $|\mathcal{K}|$ is the "easiest."

Corollary 3. *Suppose Assumption 4 holds. The maximum attainable total surplus under owner $g \in \mathcal{N}$ is*

$$TS(g) = \max_{(\pi,m)\in\mathcal{PM}(g)} \Bigg\{ \sum_{k\in\mathcal{K}} \min_{i\in\widehat{\mathcal{P}}_k(g)} P_{k,i}(\pi, m; g) \\ - \sum_{i\in\mathrm{Arm}(g)} \sum_{j\in J_i(g)\setminus\{i\}} \Big[(\lambda_{ij} - b_{ij}) \sum_{S:j\in S} \pi_{iS}\, m_{ij}^S + c_{ij} \sum_{S:j\in S} \pi_{iS}\, (1 - m_{ij}^S) \Big] \Bigg\},$$

where

$$P_{k,i}(\pi, m; g) \equiv \sum_{S\subseteq J_i(g)} \pi_{iS}\, \rho_{k,i}^S, \qquad \rho_{k,i}^S \equiv \min_{j\in\mathcal{Q}_{k,i}(g)} m_{ij}^S \quad (\min \varnothing \equiv 1).$$

Proof. Under Assumption 4, for each i we have $Q_{1,i}(g) \supseteq Q_{2,i}(g) \supseteq \cdots$. Therefore,

$$\rho_{1,i}^S = \min_{j\in Q_{1,i}(g)} m_{ij}^S \;\leqslant\; \min_{j\in Q_{2,i}(g)} m_{ij}^S = \rho_{2,i}^S \;\leqslant\; \cdots \;\leqslant\; \rho_{|K|,i}^S.$$

Hence $P_{k,i}(\pi, m; g) = \sum_S \pi_{iS}\rho_{k,i}^S$ is non-decreasing in k.

Step 1: Nested Regions of Success. In the construction of an optimal decision rule, this monotonicity implies that the success regions are nested:

$$\mathcal{S}_{1,i} \subseteq \mathcal{S}_{2,i} \subseteq \cdots \subseteq \mathcal{S}_{|K|,i}, \qquad \mathrm{Leb}(\mathcal{S}_{k,i}) = P_{k,i}(\pi, m; g).$$

As a result, the only nonempty cells $\Delta_{i,T}$ correspond to *upper sets* of the form

$$T^{(k)} \equiv \{k, k+1, \ldots, |K|\} \quad (k = 1, \ldots, |K|), \qquad T^{(|K|+1)} \equiv \varnothing,$$

with

$$\Delta_{i,T^{(k)}} = \mathcal{S}_{k,i} \setminus \mathcal{S}_{k-1,i}, \qquad \Delta_{i,\varnothing} = \mathcal{S}_{|K|,i}^c,$$

where $\mathcal{S}_{0,i} \equiv \varnothing$. Therefore the type marginals are

$$p_{i,T^{(k)}}(\pi, m; g) = \mathrm{Leb}(\Delta_{i,T^{(k)}}) = P_{k,i} - P_{k-1,i}, \qquad p_{i,\varnothing}(\pi, m; g) = 1 - P_{|K|,i},$$

with the convention $P_{0,i} \equiv 0$.

Step 2: Upper Bound on MMOT. Let γ be any feasible coupling in (A.27), i.e. with marginals $\beta_{iT}(\pi, m; g)$. For any project k, the MMOT objective involves the probability that

all required parties succeed in project k:

$$\mathbb{P}_\gamma\big(k \in T_i \ \forall i \in \widetilde{\mathcal{P}}_k(g)\big) \ \leqslant \ \min_{i \in \widetilde{\mathcal{P}}_k(g)} \mathbb{P}_\gamma(k \in T_i).$$

Under the fixed marginal of party i,

$$\mathbb{P}_\gamma(k \in T_i) = \sum_{T:\, k \in T} \beta_{iT}(\pi, m; g) = P_{k,i}(\pi, m; g),$$

because $k \in T_i$ is the event $(\eta_i, \xi_i) \in \mathcal{S}_{k,i}$, whose area is $P_{k,i}$. Therefore,

$$\mathbb{P}_\gamma\big(k \in T_i \ \forall i \in \widetilde{\mathcal{P}}_k(g)\big) \ \leqslant \ \min_{i \in \widetilde{\mathcal{P}}_k(g)} P_{k,i}(\pi, m; g).$$

Multiplying by V_k, summing over k, and taking the maximum over feasible γ yields

$$\mathrm{W}(\pi, m; g) \ \leqslant \ \sum_{k \in \mathcal{K}} V_k \min_{i \in \widetilde{\mathcal{P}}_k(g)} P_{k,i}(\pi, m; g).$$

Step 3: Attainability of the Upper Bound on MMOT. Let $V \sim \mathrm{Unif}[0,1]$ be a *common* random variable shared across all parties. For each party i, define its type as the ladder determined by the points $(P_{k,i})_k$:

$$T_i(V) \equiv \{k \in K : \ V \leqslant P_{k,i}(\pi, m; g)\}.$$

Because $P_{k,i}$ is non-decreasing in k, $T_i(V)$ is always an upper set. Moreover,

$$\mathbb{P}\big(T_i(V) = T^{(k)}\big) = \mathbb{P}\big(P_{k-1,i} < V \leqslant P_{k,i}\big) = P_{k,i} - P_{k-1,i} = \beta_{i,T^{(k)}}(\pi, m; g),$$

and $\Pr(T_i(V) = \varnothing) = 1 - P_{|K|,i} = \beta_{i,\varnothing}(\pi, m; g)$. Hence the induced joint law of $(T_i(V))_{i \in \mathcal{I}(g)}$ is a feasible coupling for (A.27).

Under this coupling,

$$k \in T_i(V) \ \forall \ i \in \widetilde{\mathcal{P}}_k(g) \quad \Longleftrightarrow \quad V \leqslant \min_{i \in \widetilde{\mathcal{P}}_k(g)} P_{k,i}(\pi, m; g),$$

so

$$\mathbb{P}\big(k \in T_i(V) \ \forall i \in \widetilde{\mathcal{P}}_k(g)\big) = \min_{i \in \widetilde{\mathcal{P}}_k(g)} P_{k,i}(\pi, m; g).$$

Therefore, the MMOT objective value achieved by this coupling equals

$$\sum_{k\in\mathcal{K}} V_k \min_{i\in\widetilde{\mathcal{P}}_k(g)} P_{k,i}(\pi, m; g),$$

which matches the upper bound from Step 2. Hence

$$\mathrm{W}(\pi, m; g) = \sum_{k\in\mathcal{K}} V_k \min_{i\in\widetilde{\mathcal{P}}_k(g)} P_{k,i}(\pi, m; g).$$

□